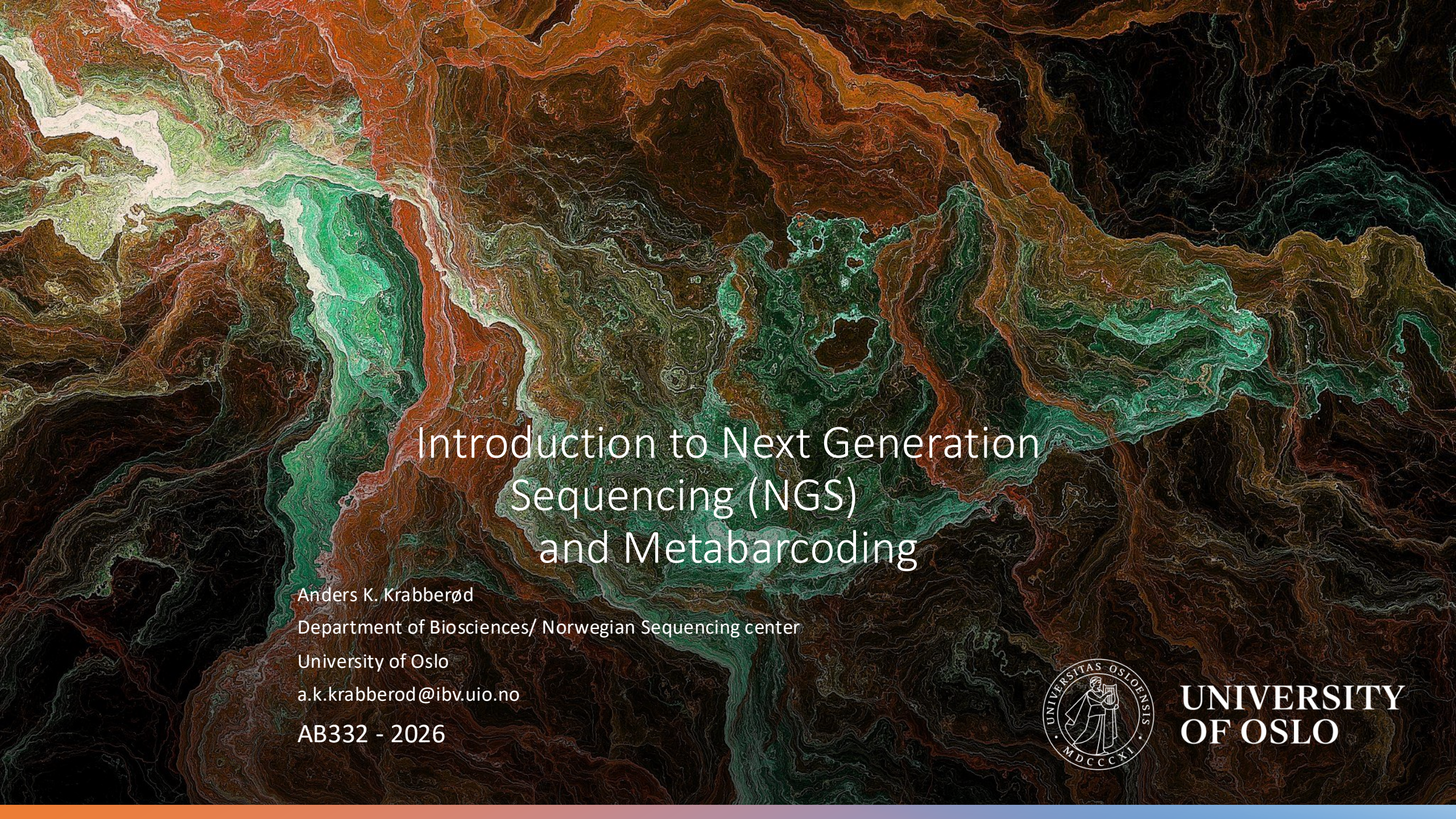


Introduction to Next Generation Sequencing (NGS) and Metabarcoding

Anders K. Krabberød
Department of Biosciences/ Norwegian Sequencing center
University of Oslo
a.k.krabberod@ibv.uio.no

AB332 - 2026

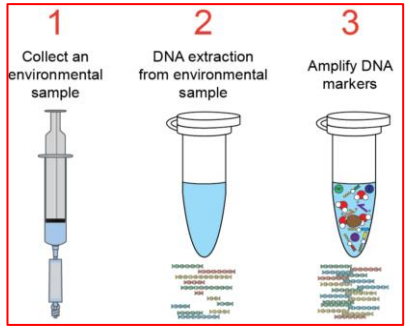
UNIVERSITY OF OSLO

Some important terms

- Metabarcoding – The use of DNA markers to analyze biodiversity and species identification from environmental samples.
- Amplicons - DNA fragments generated through polymerase chain reaction (PCR)
- OTUs (and ASVs) - Groupings of closely related sequences used as proxies for species (Operational Taxonomic Unit, Amplicon Sequence Variants)
- High-throughput Sequencing (HTS) - A broad range of technologies that enable the rapid sequencing of large volumes of DNA or RNA
- Next Generation Sequencing (NGS) - A specific type of HTS technology that allows for massively parallel sequencing of DNA (or RNA)
- Third-generation Sequencing - A newer generation of sequencing technologies (e.g., PacBio, Oxford Nanopore) that read long, single molecules of DNA or RNA in real-time

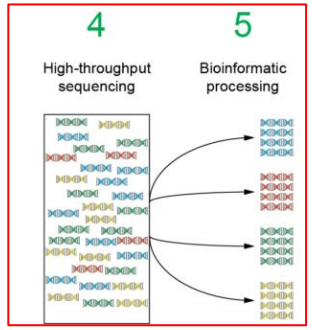
Metabarcoding

Fieldwork/labwork/
DNA extraction etc



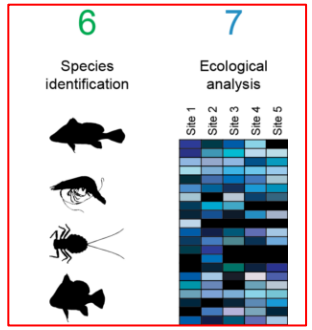
Sequencing /
Bioinformatics

Tuesday

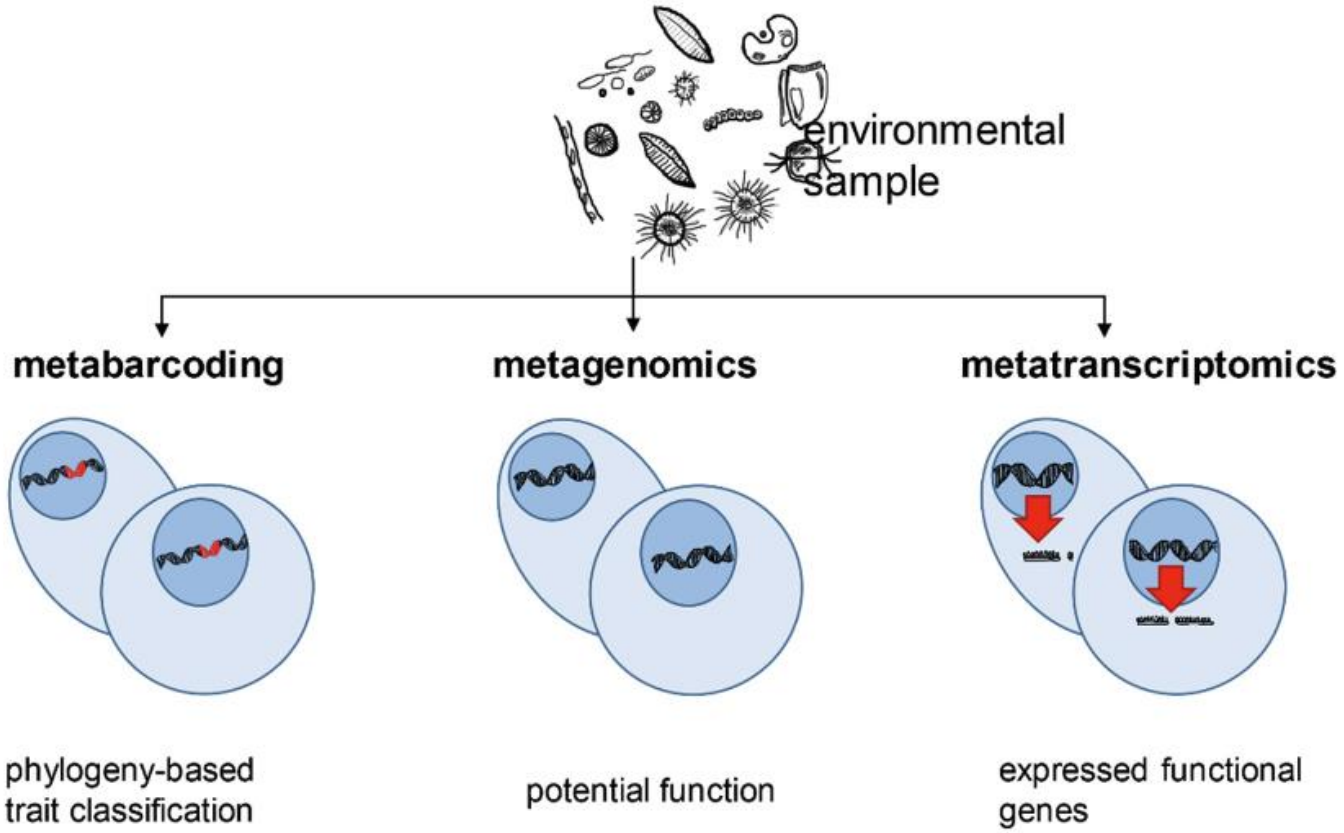


Ecological
analysis

Computer Labs
Tuesday and Wednesday



Abdelfattah et al.,2018

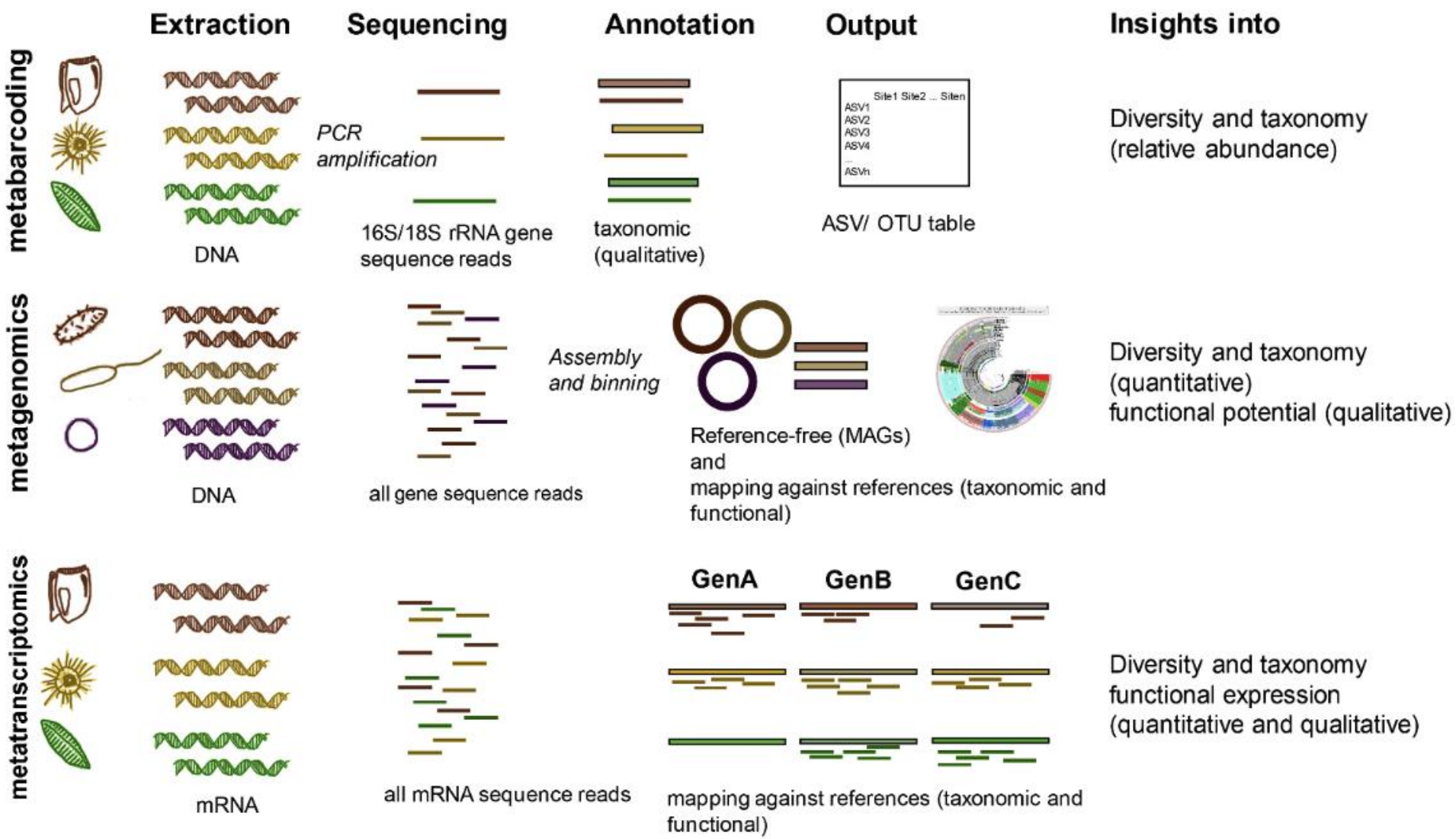


Who is present?

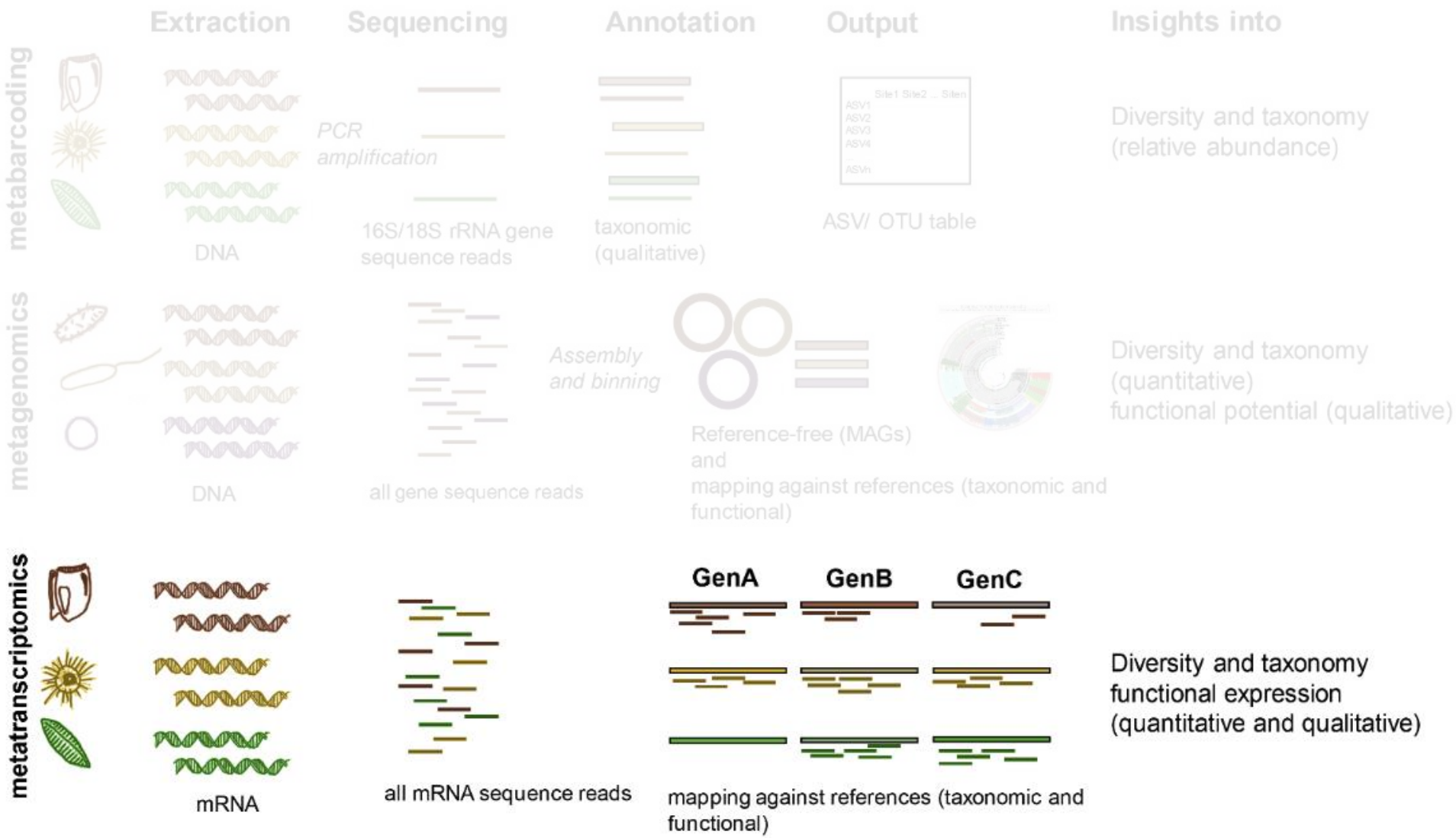
Who, and potential functionality

Who is active, and which genes?

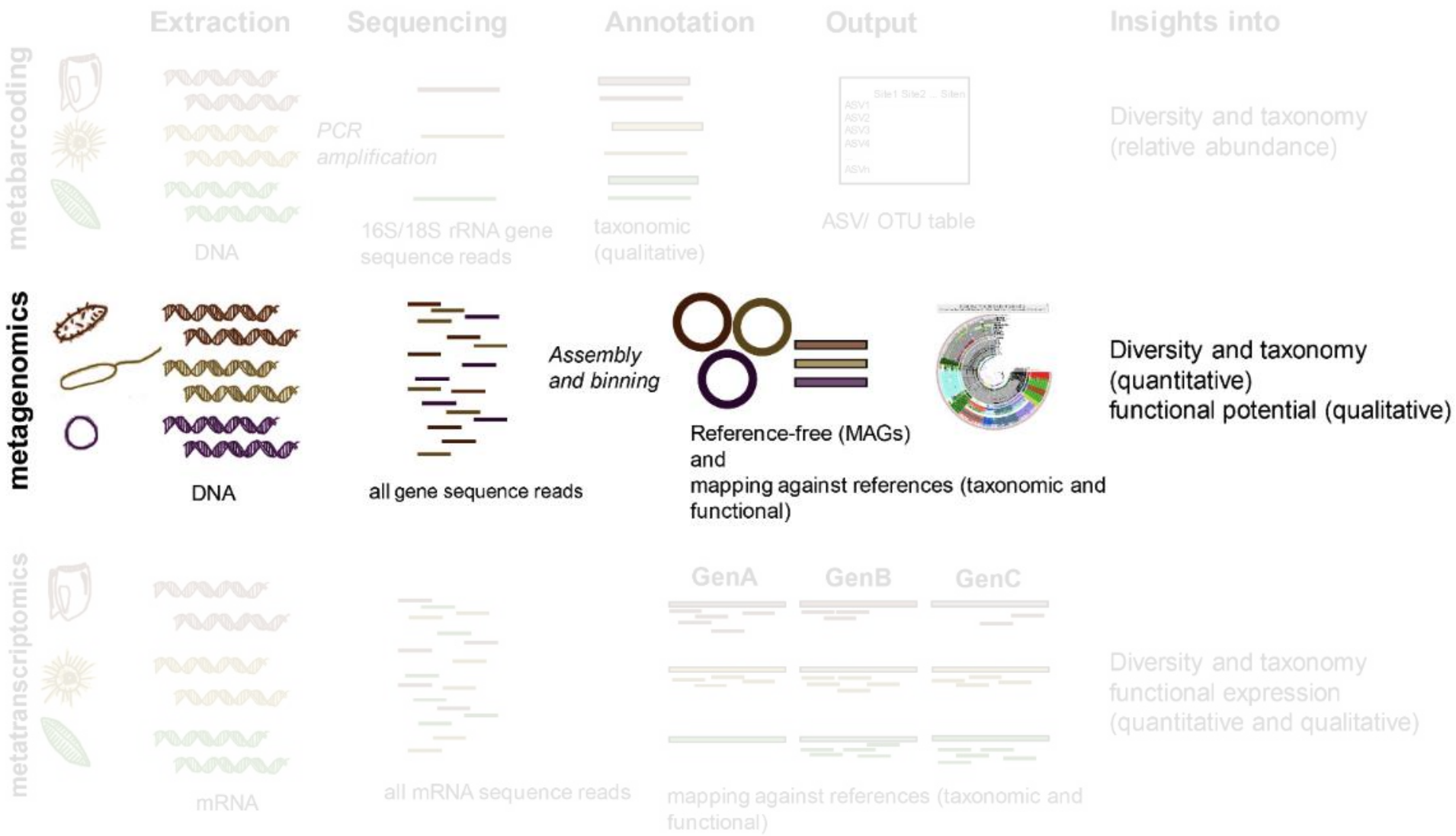
Hoerstmann et al 2022



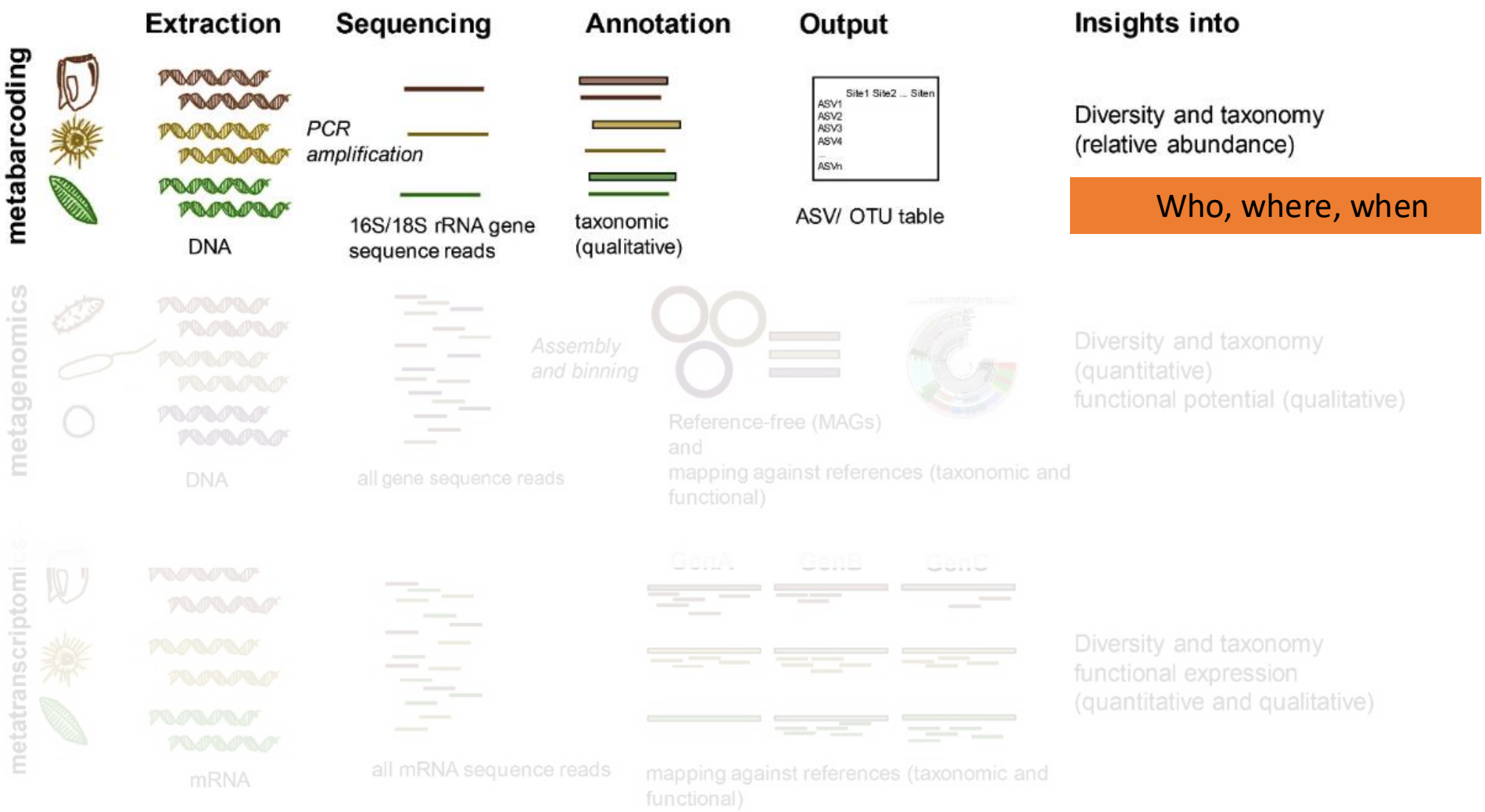
Hoerstmann et al 2022



Hoerstmann et al 2022



Hoerstmann et al 2022



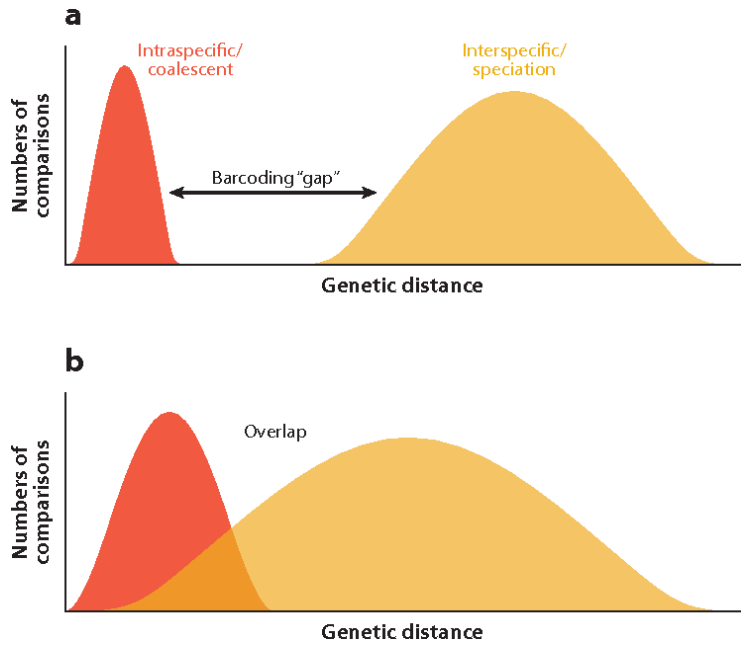
Hoerstmann et al 2022

Metabarcoding

- Typical research questions:
  - Who is present, where, and when?
  - Richness: How many taxa/species/OTUs (alpha/gamma diversity)?
    - **OTU:** Operational taxonomic unit -> the group of organisms currently being studied (Sokal & Sneath 1957), a modern take would be that OTUs are pragmatic proxies for "species".
  - Compositional differences (beta diversity)?
  - Which processes and drivers are shaping the communities?
  - Co-occurrence patterns (possible interactions)

Markers in DNA metabarcoding

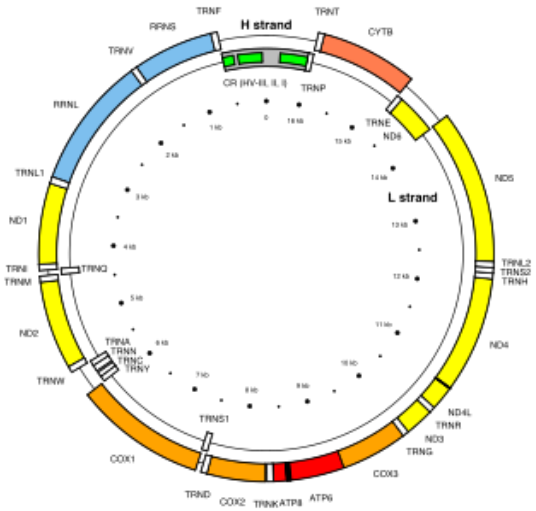
- The ideal marker should:
  - Have primer sites that are shared by all target organisms
  - Be easy to amplify in PCR
  - Be of appropriate length for efficient *amplification* and *sequencing*
  - Be of similar length for all target organisms
  - No intragenomic variation (i.e. no paralogs)
  - Similar number of copies
  - Have high interspecific variation
  - Have low intraspecific variation
  - Be possible to align (not always required)
- Likely no marker will meet all these requirements!

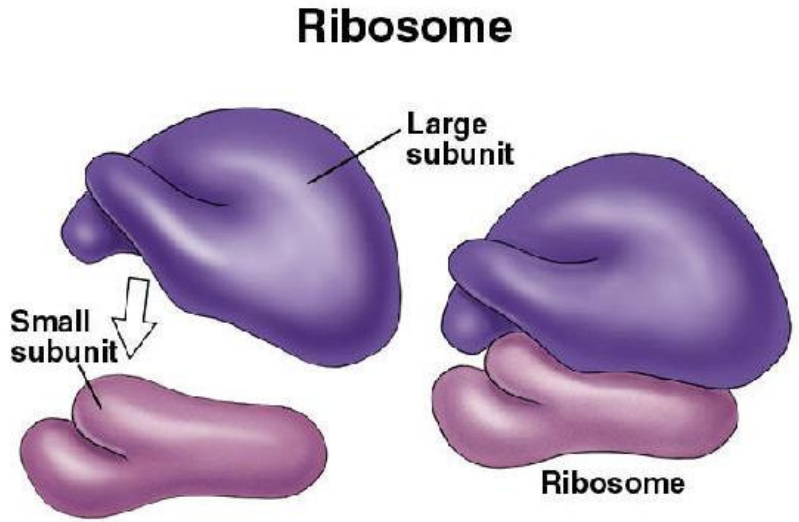


Markers used in DNA metabarcoding

- Standard markers (<500 bp):
  - 18S: Eukaryotes
  - 16S: Bacteria/archaea
  - ITS: Fungi & plants
  - COI: Metazoa
  - *Rbc*L: Plants
  - *trn*L: Plants

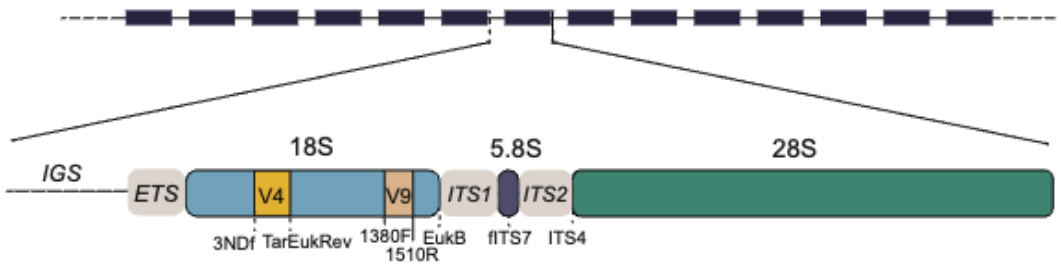






Ribosomal operon

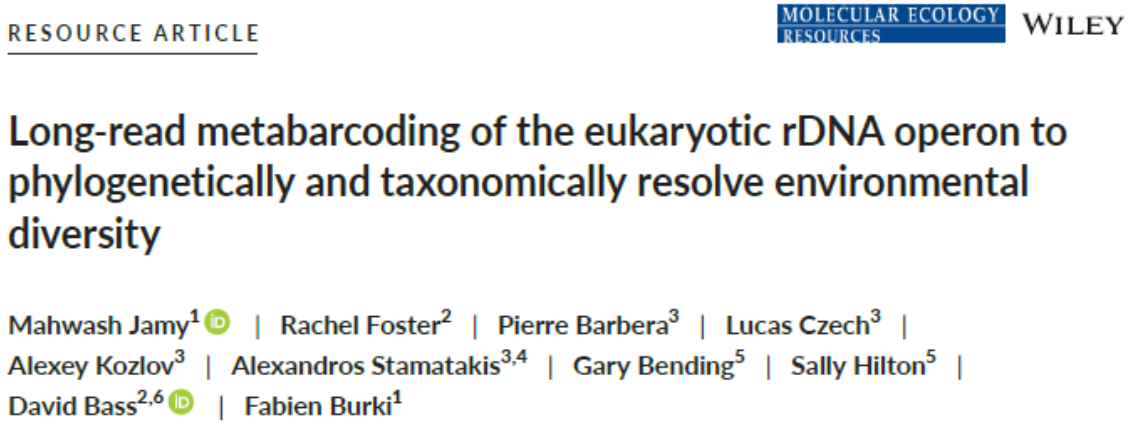
- Often shortened rRNA or rDNA
- 16S – 5S - 23S in prokaryotes
- 18S – 5.8S -28S in eukaryotes
  - S stands for Svedberg units; a unit of molecular size determined by centrifugation.
- Present across the Tree of Life!
- The full length is in the range of 5000-7000 (but with a lot of variation)
- Typically for Illumina sequencing a region of 300-450bp is used (e.g. V4).



Long-read metabarcoding

- Sequencing the full operon is possible with the development of sequencing technologies
- Stronger phylogenetic signal

- Comes with extra challenges
  - Harder to amplify longer regions
  - More chimeric sequences
  - Lower sequencing depth



RESOURCE ARTICLE

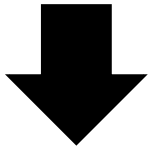
MOLECULAR ECOLOGY RESOURCES WILEY

Long-read metabarcoding of the eukaryotic rDNA operon to phylogenetically and taxonomically resolve environmental diversity

Mahwash Jamy[1] | Rachel Foster[2] | Pierre Barbera[3] | Lucas Czech[3] | Alexey Kozlov[3] | Alexandros Stamatakis[3,4] | Gary Bending[5] | Sally Hilton[5] | David Bass[2,6] | Fabien Burki[1]

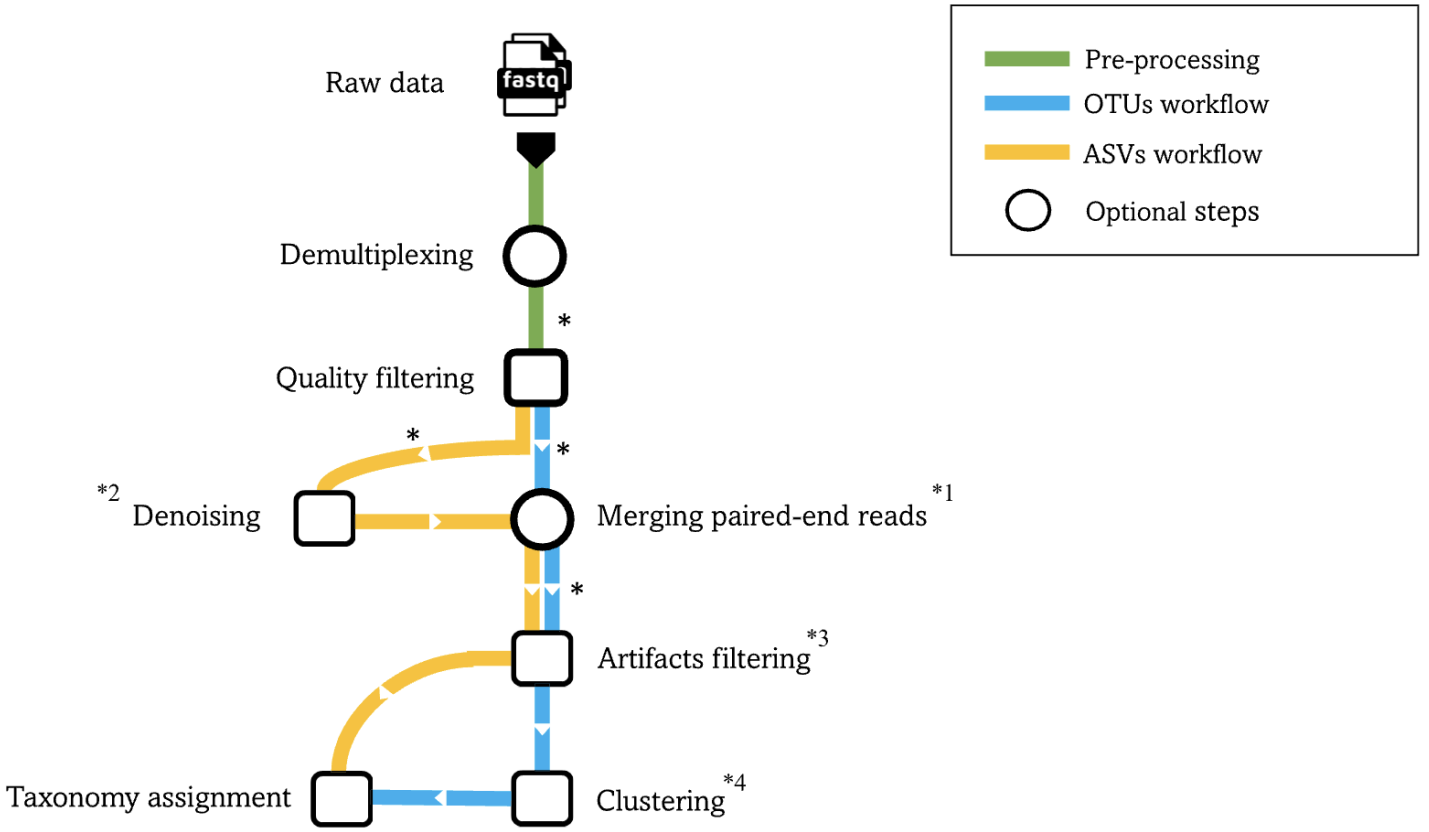
How conserved/variable is the marker?

- 18S (and 16S): (Relatively) Low variability, low intraspecific variation, low interspecific variation
- ITS: High variability, high intraspecific variation, high ‘interspecific’ variation
- COI, high intraspecific variation
- Sequencing platforms and PCR also introduce errors



- Impact how the bioinformatics analyses should be conducted

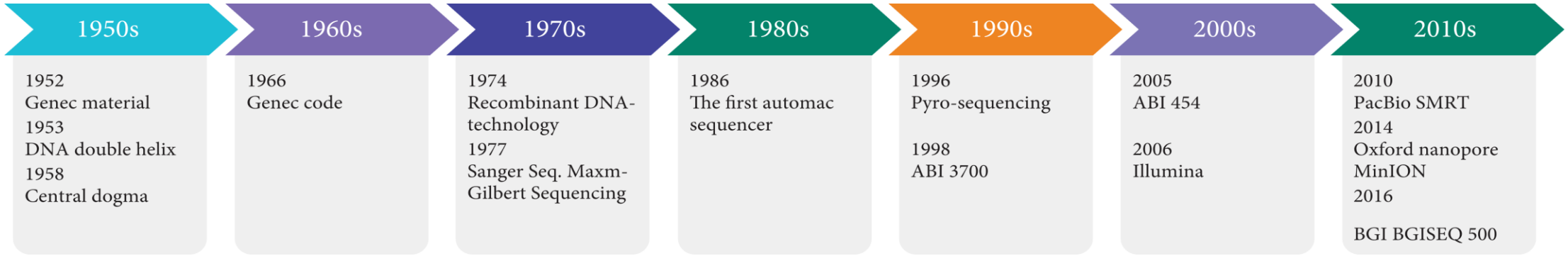
Metabarcoding bioinformatics workflow



Hakimzadeh et al 2024

A short history of Sequencing

- Sequencing: determining the order of nucleotide bases in a string of DNA (or RNA)
- The development started in the 1950's after Watson and Crick described the structure of DNA
- The early methods were cumbersome (and dangerous) using radioactive material and adding individual nucleotides to a reaction one by one.
- The last few years the development has been phenomenal!



A short history of Sequencing

- F. Sanger et al. 1977
  - Short fragments
  - 15-200 nucleotides
  - Sloooooow process

Proc. Natl. Acad. Sci. USA
Vol. 74, No. 12, pp. 5463–5467, December 1977
Biochemistry

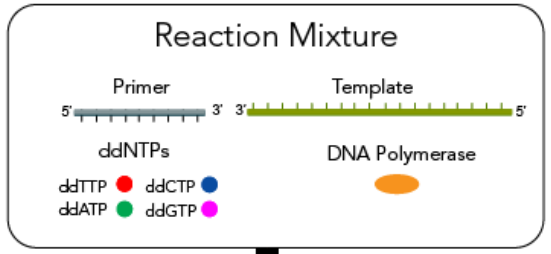
DNA sequencing with chain-terminating inhibitors

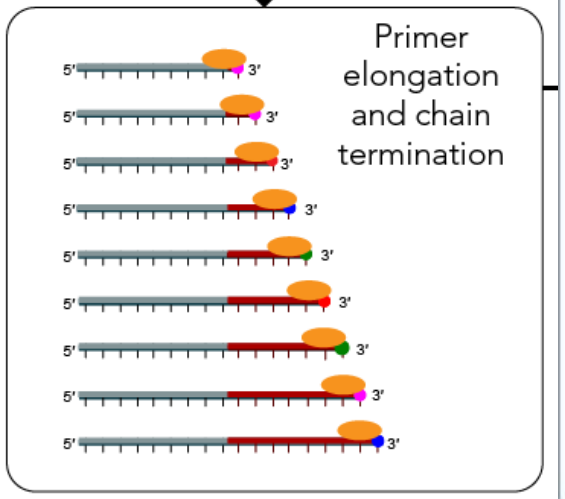
(DNA polymerase/nucleotide sequences/bacteriophage ϕX174)

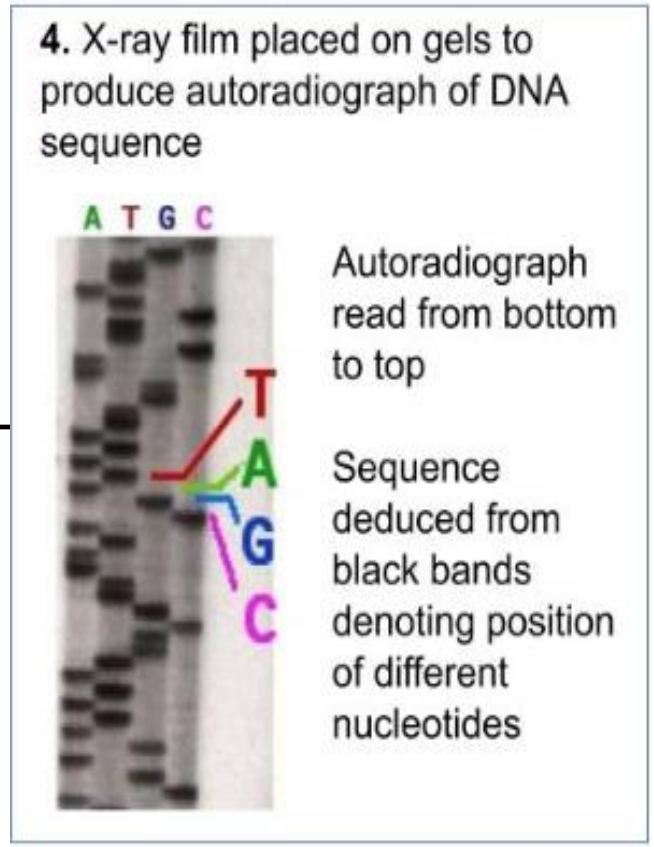
F. SANGER, S. NICKLEN, AND A. R. COULSON

Medical Research Council Laboratory of Molecular Biology, Cambridge CB2 2QH, England

Contributed by F. Sanger, October 3, 1977

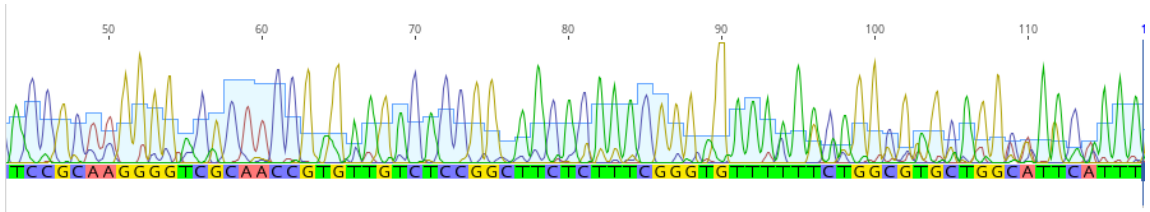


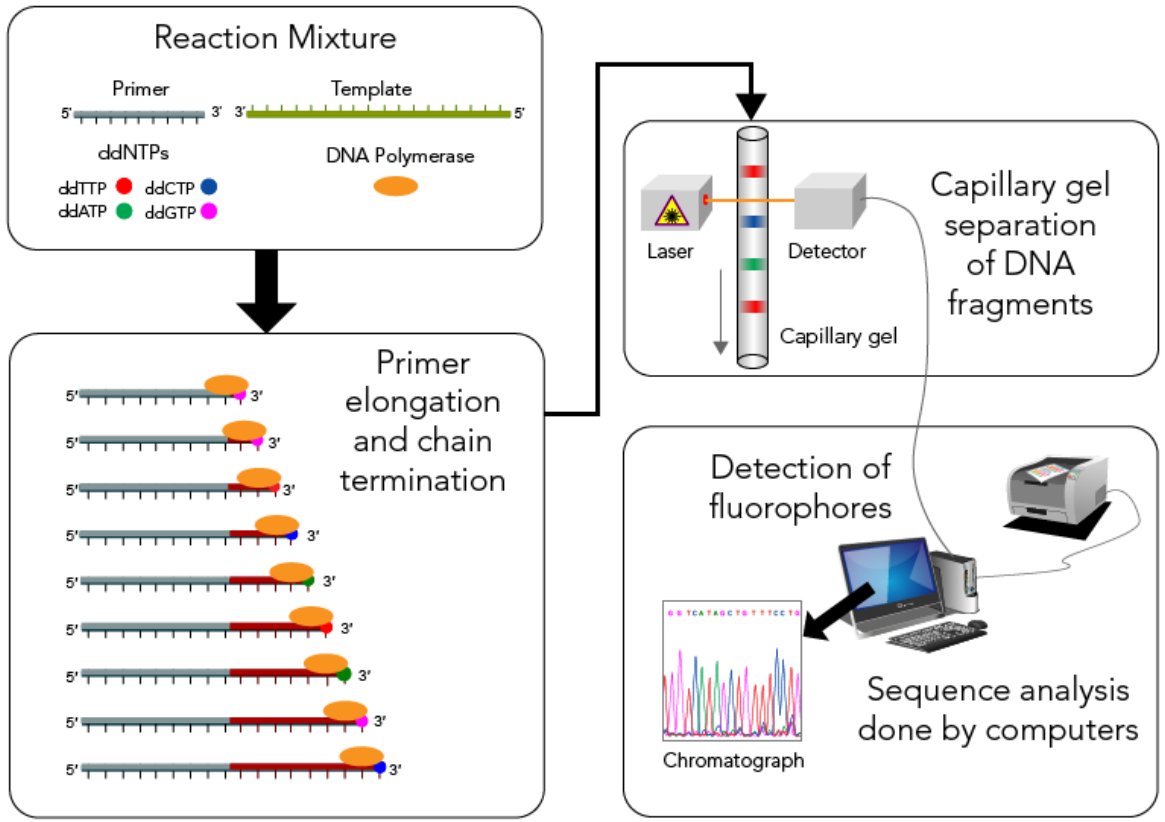




A short history of Sequencing

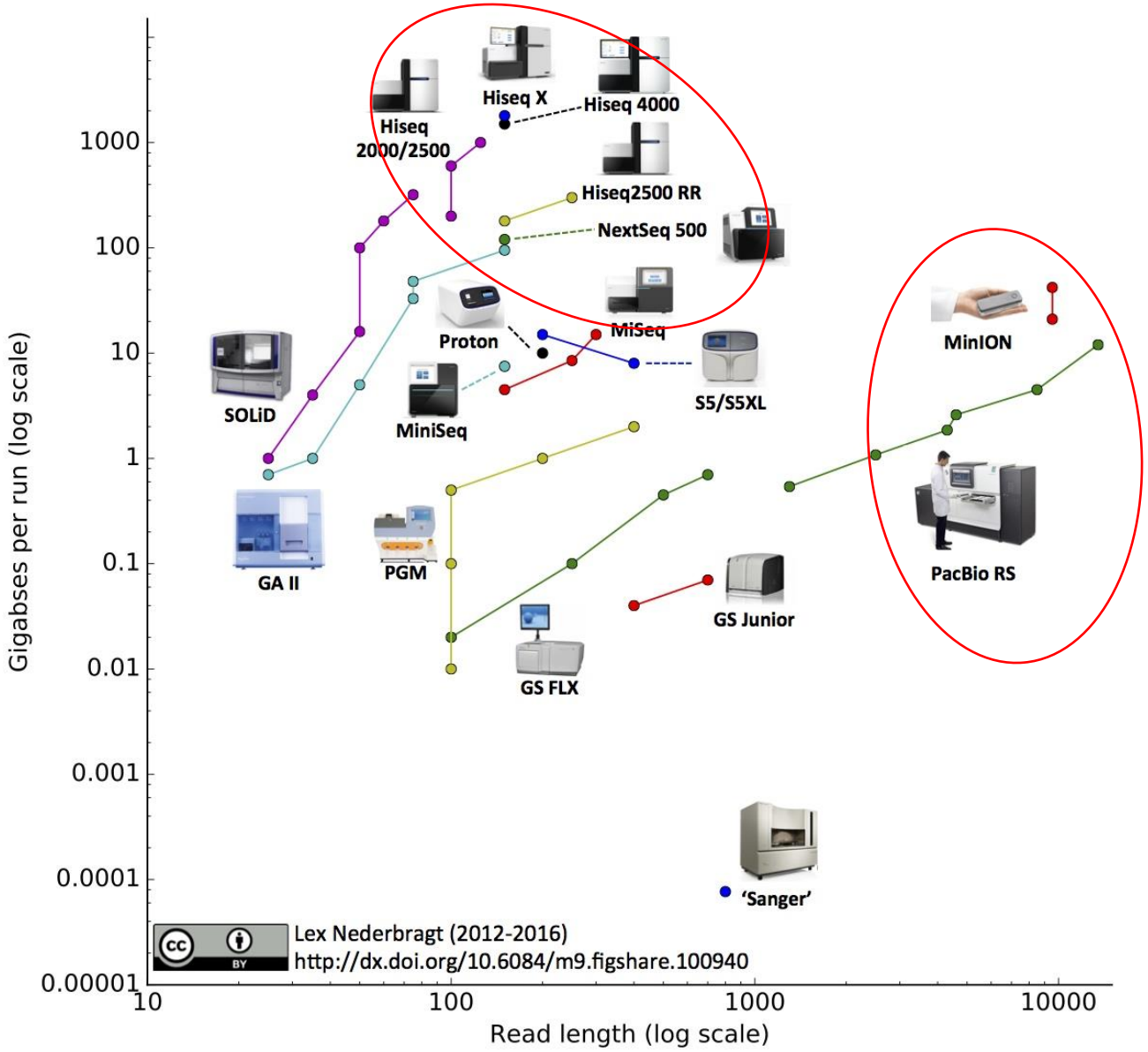
- F. Sanger et al. 1977
  - Short fragments
  - 15-200 nucleotides
  - Sloooooow process
- Applied Biosystems automating the process in the late 80's early 90' with capillary electrophoresis, fluorescent dyes, and lasers.
- "First generation sequencing"
  - 500—1000 nucleotides
  - Slow, but much faster than manual
  - The main technique for the human genome project
  - Sequencing 3 gigabases took 10 years
  - Still used, since it is very high quality and cheap (if you only want to look at a handful of sequences)





High Throughput Sequencing (HTS)

- Early 2000's and onwards
- High Throughput Sequencing (HTS) is a collective term
  - Next generation sequencing (NGS)
    - Short reads, (100-300bp)
    - but generates a huge amount of reads (in the billions)
    - 454 Roche
    - **Illumina (HiSeq, MiSeq, NovaSeq, NextSeq)**
    - Ion torrent
  - Third generation sequencing
    - Longer reads, (1000-100kbp)
    - not so many as NGS, but still in the 100k or millions
    - Oxford Nanopore (MinIon, GridIon,PromethIon, etc)
    - **PacBio (Sequel, Revio)**


Gigabses per run (log scale)
1000
100
10
1
0.1
0.01
0.001
0.0001
0.00001
10
100
1000
10000
Read length (log scale)
Hiseq 2000/2500
Hiseq X
Hiseq 4000
Hiseq2500 RR
NextSeq 500
Proton
MiSeq
SOLiD
MiniSeq
S5/S5XL
MinION
GA II
PGM
PacBio RS
GS Junior
GS FLX
'Sanger'
Lex Nederbragt (2012-2016)
http://dx.doi.org/10.6084/m9.figshare.100940

Illumina

- “Sequencing by synthesis”
- Short fragments
  - 150-300 bp in pairs
- Low error rate (0.1% - 0.5%)
- MiSeq output (2*300bp):
  - 25 million reads (15Gb)
- NextSeq (2*300 bp):
  - 1.2 billion reads (360Gb)
- NovaSeq 6000
  - 20 billion reads (6Tb)
- Other platforms exist



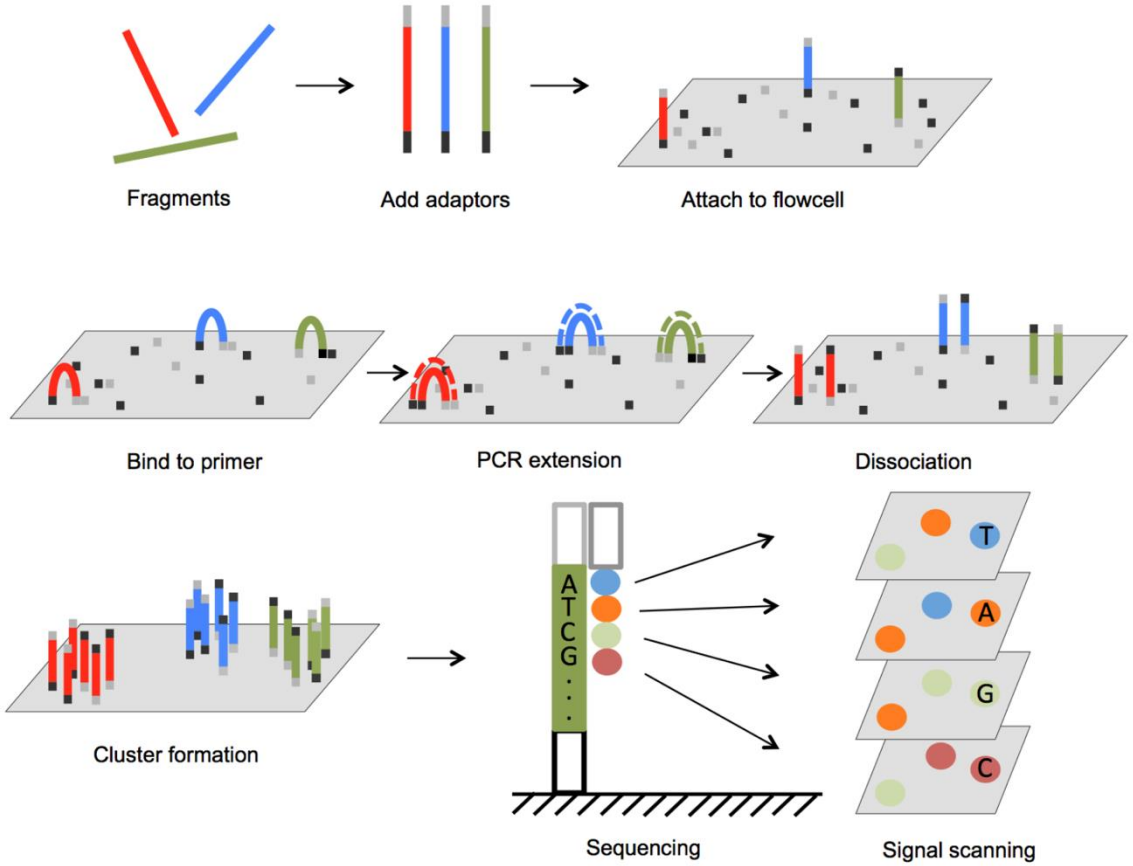


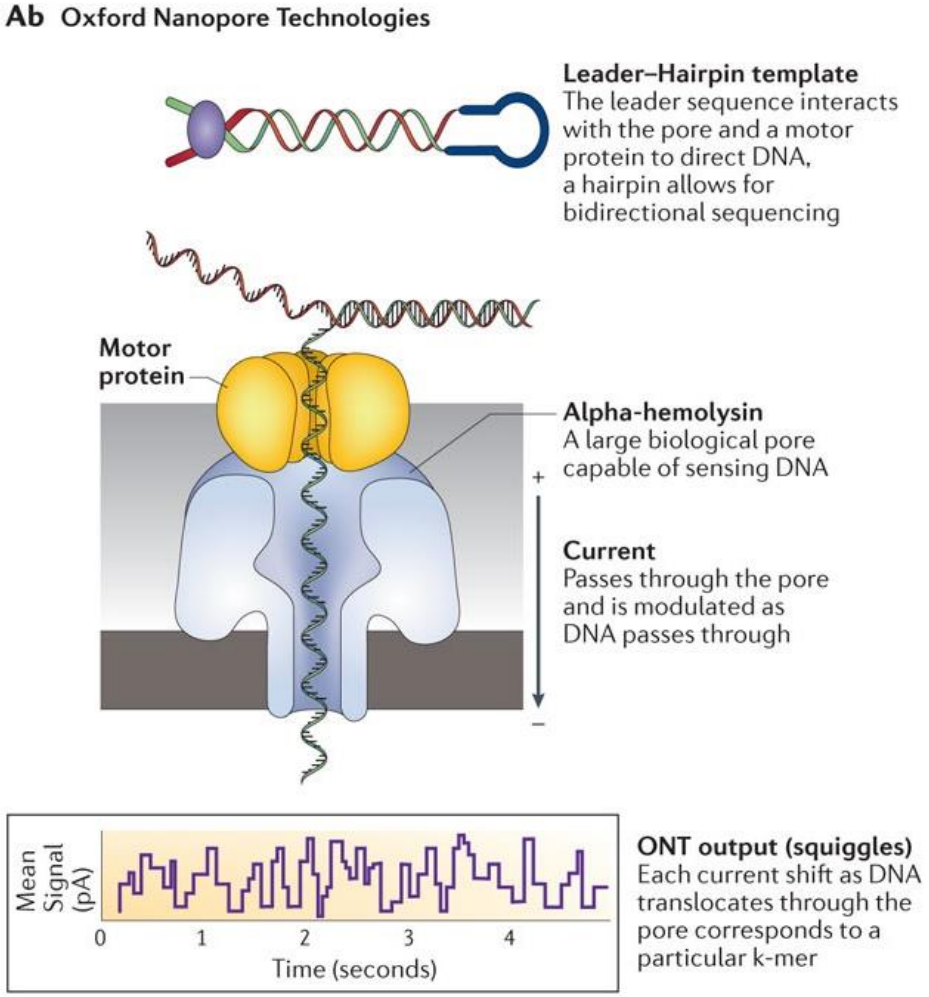
Figure 1: Principle of the illumina sequencing by synthesis (SBS) technology (Lu et al., 2016)

https://www.youtube.com/watch?v=fCd6B5HRaZ8&t=1s

Oxford Nanopore

- Long to very long reads (10kb -100kb)
- Higher error rate, but it is improving
- Lower output than Illumina
  - MinION (50Gb)
  - PromethION (290Gb)
- Realtime sequencing
- Portable





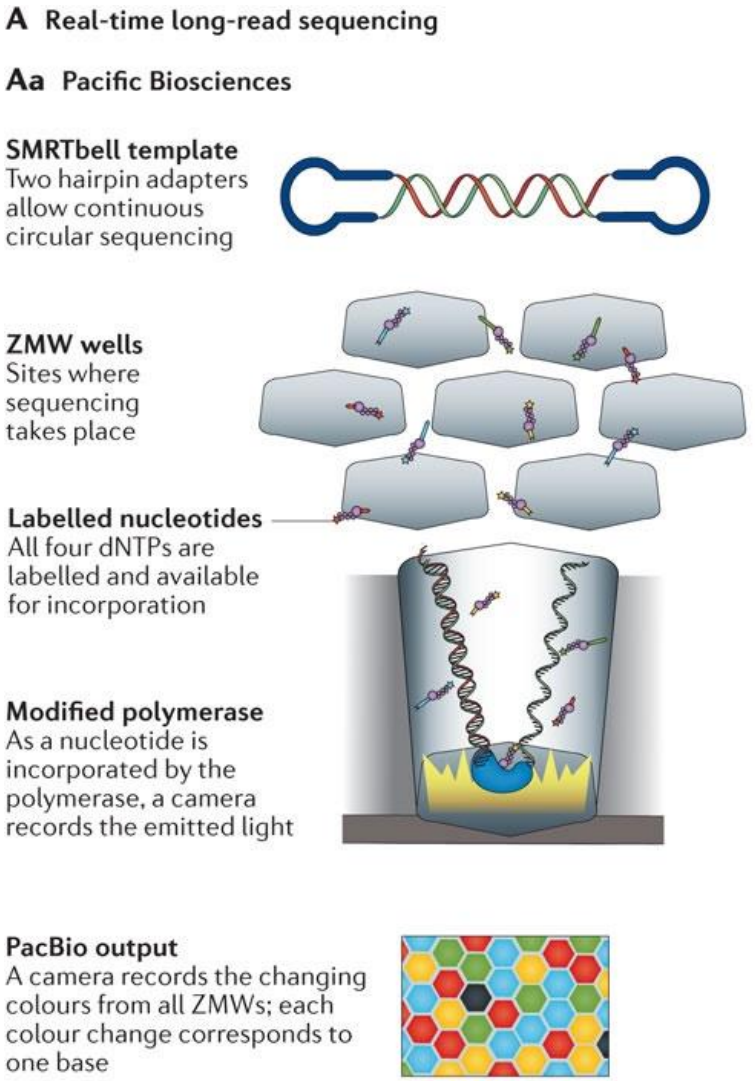
https://www.youtube.com/watch?v=RcP85JHLmnI

Goodwin, et al 2016

PacBio

- SMRT-sequencing
  - Single-molecule Real Time
- Long reads (~15kb)
- Low error rate (0.1%)
- High output
  - Theoretical output:
  - Sequel II (up to 8M reads, 120 Gb)
  - Revio (up to 23M reads, 3Tb)
  - (Real output is ~75% of this)

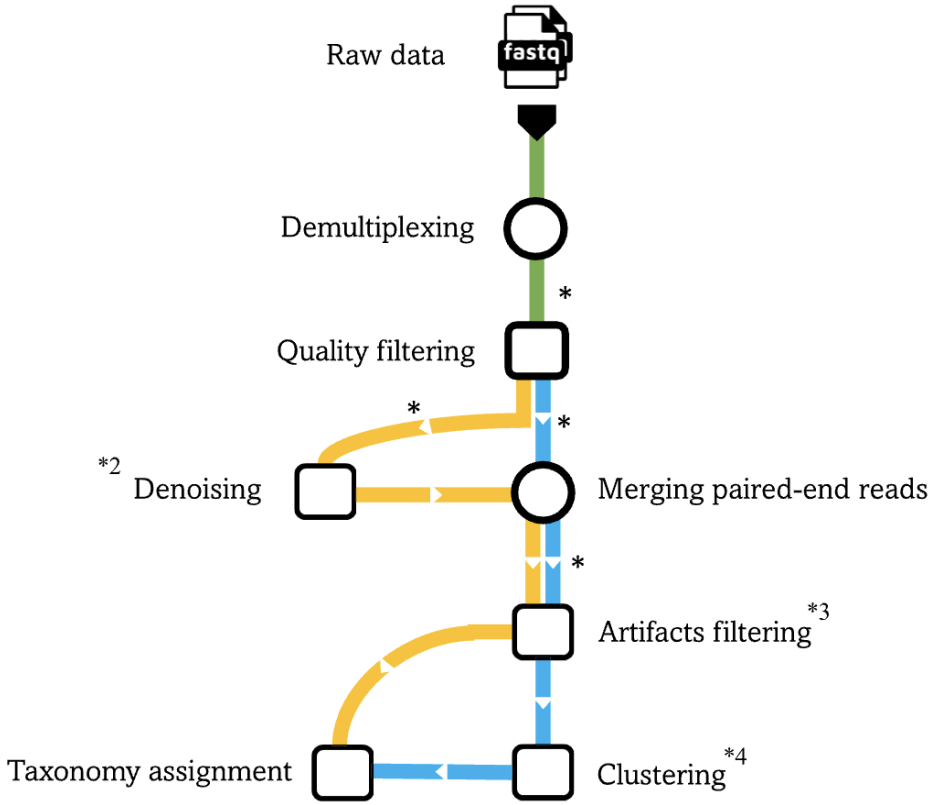




https://www.youtube.com/watch?v=NHCJ8PtYCFc

Goodwin, et al 2016

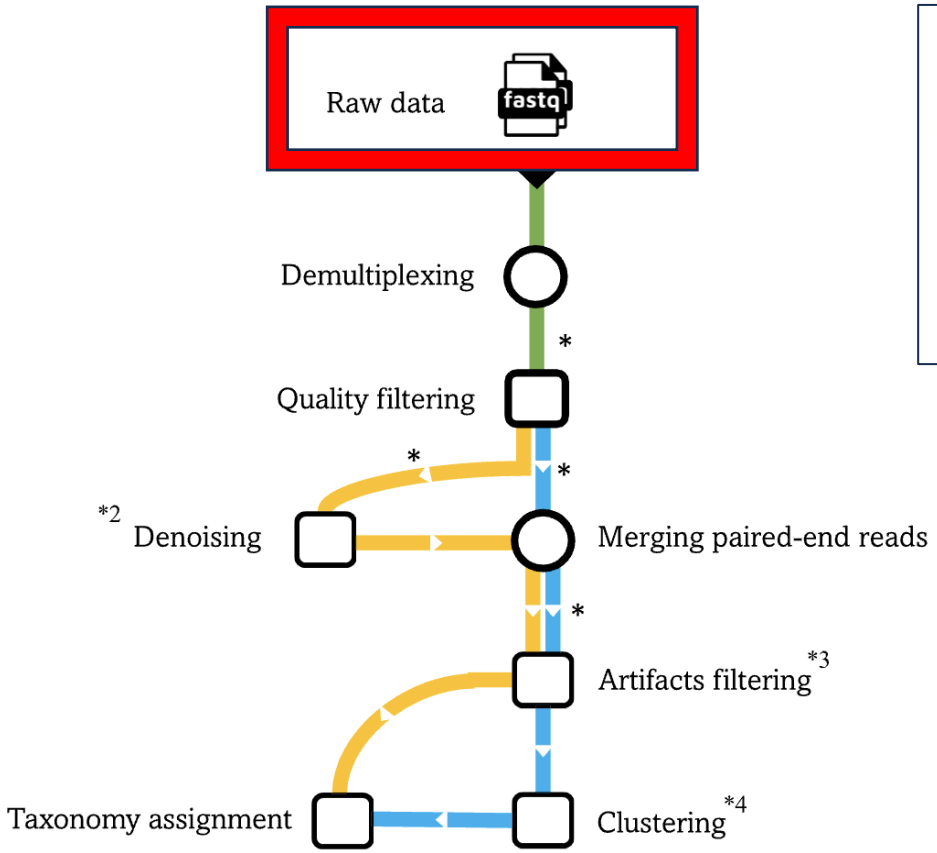
Metabarcoding bioinformatics workflow



Hakimzadeh et al 2024

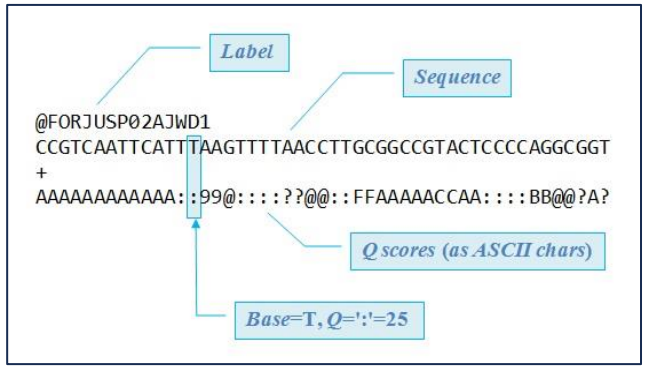
- These are the main steps when working with HTS data
- The details will vary depending on the sequencing technology used, the community under study, and the scientific answers being asked
- Some pipelines are built to do all steps for you automatically (Qiime, LotuS)
- Or you need to pick the relevant tool for your data

Metabarcoding: common file format



Hakimzadeh et al 2024

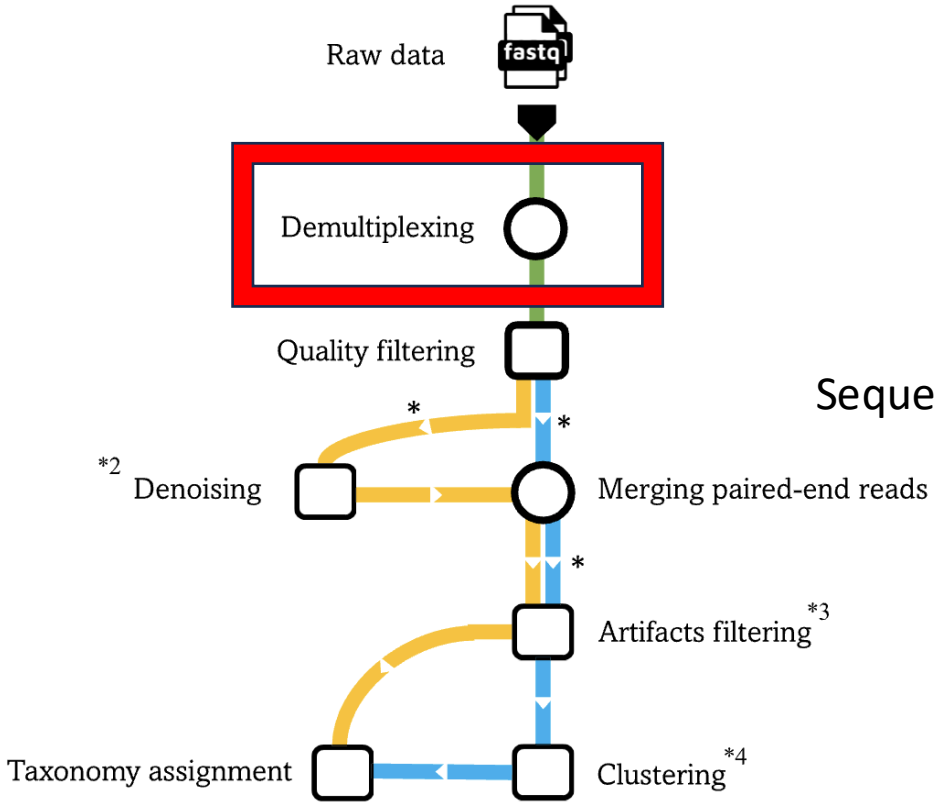
The *fastq* format

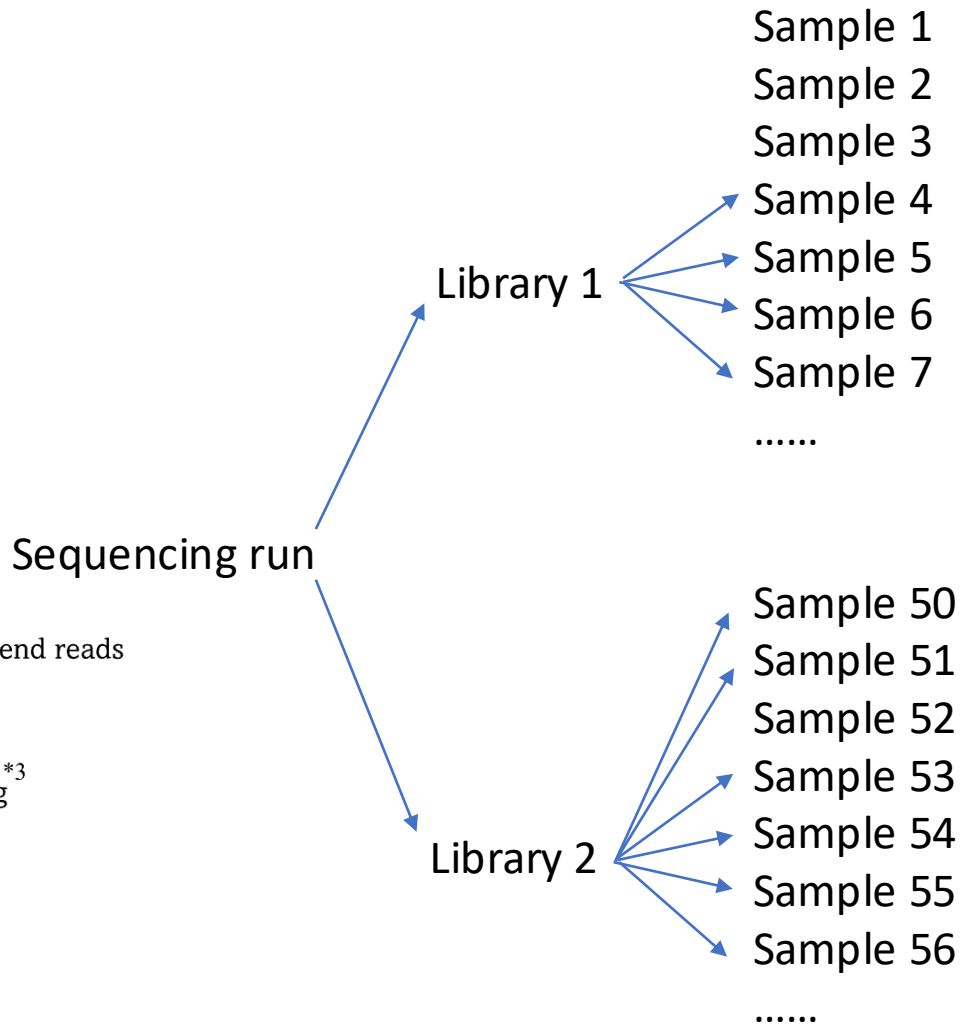


The quality is in phred score

- 1-60, coded in ASCII characters
- 20 is 99% accuracy, 30 is 99.9%
- For a modern interpretation: https://fastqe.com/

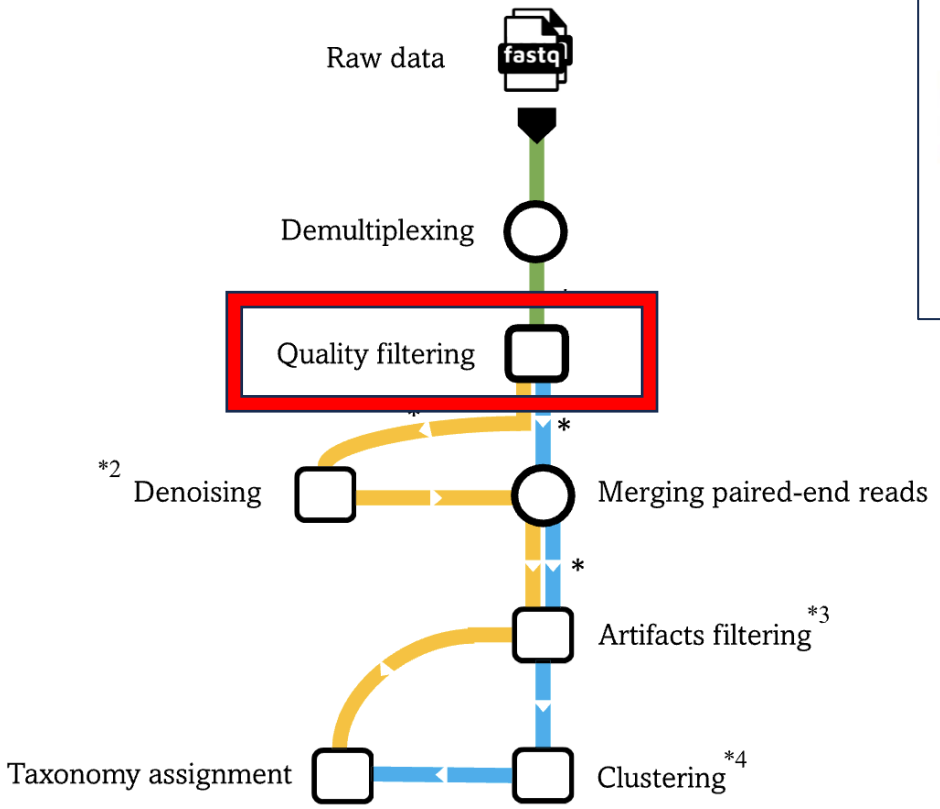
Metabarcoding: Demultiplexing





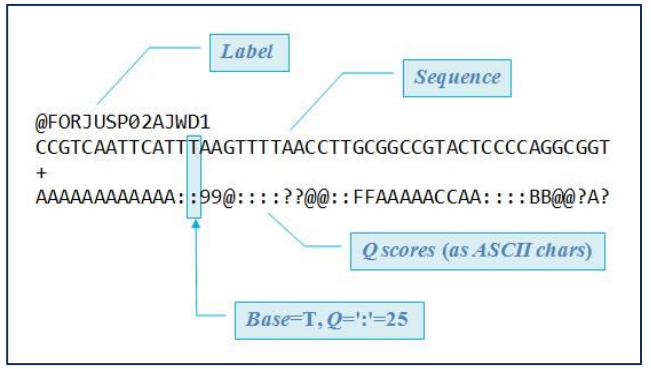
Hakimzadeh et al 2024

Metabarcoding: quality filter



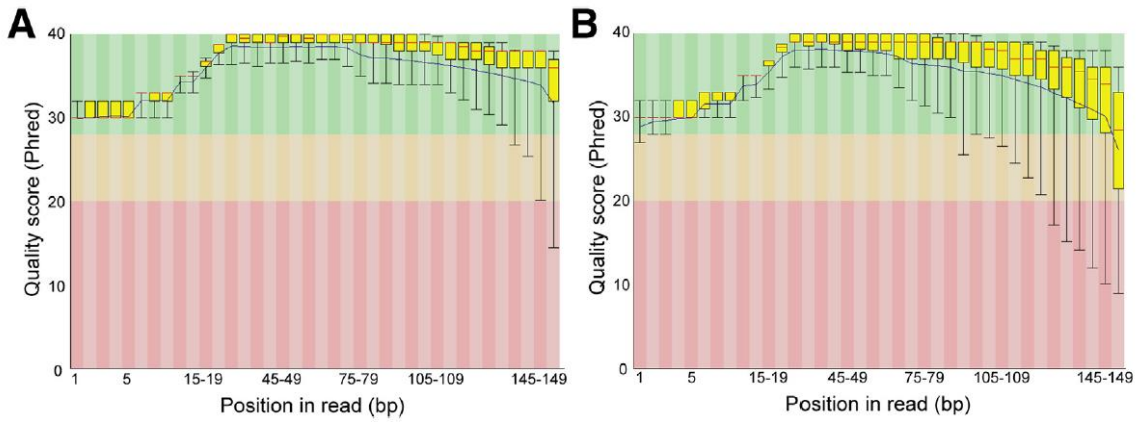
Hakimzadeh et al 2024

The *fastq* format



The quality is in phred score

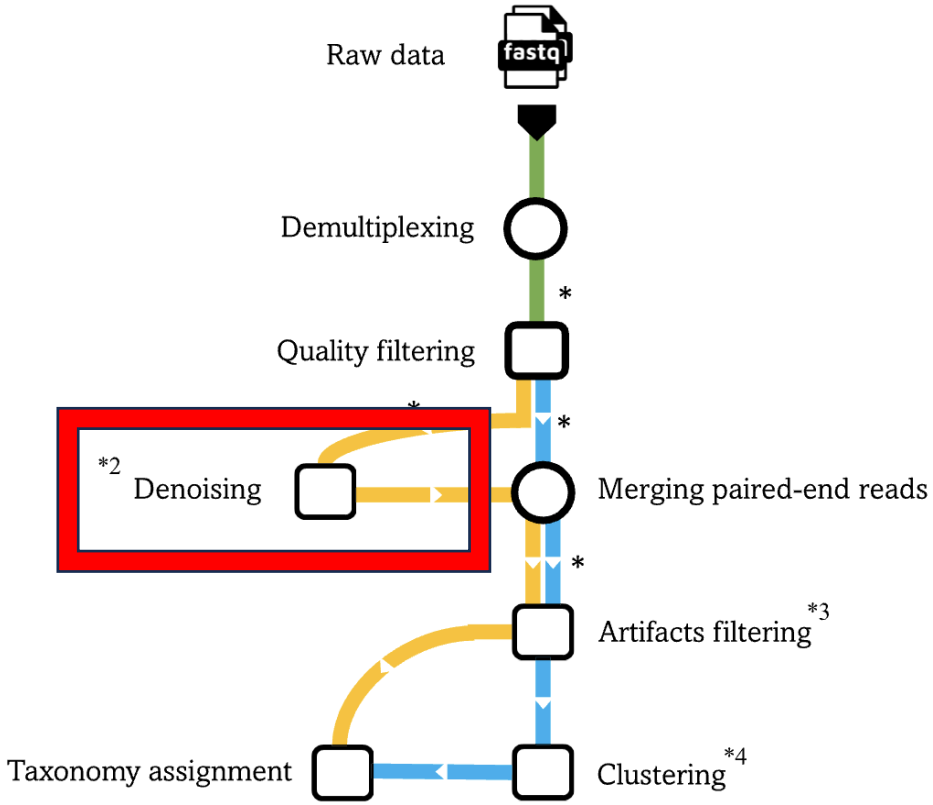
- 1-60, coded in ASCII characters
- 20 is 99% accuracy, 30 is 99.9%
- For a modern interpretation: https://fastqe.com/



Remove

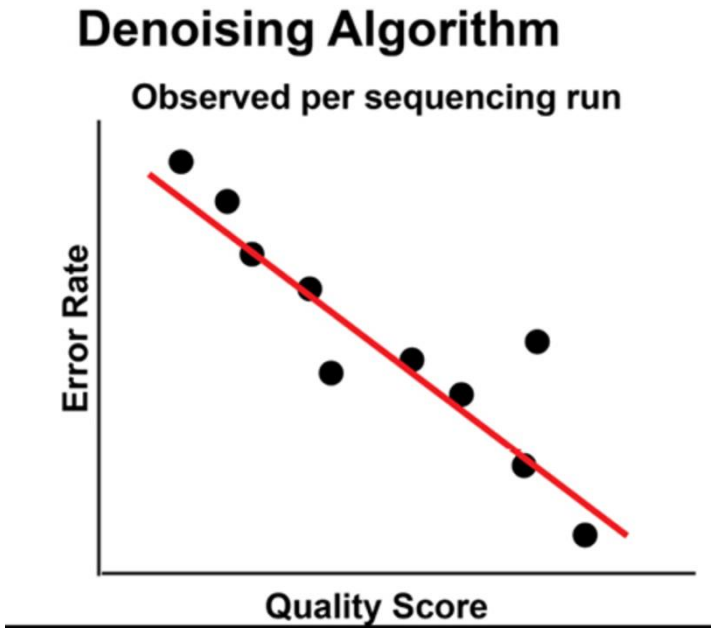
- Poor sequence quality
- Long/short sequences

Metabarcoding: Denoising



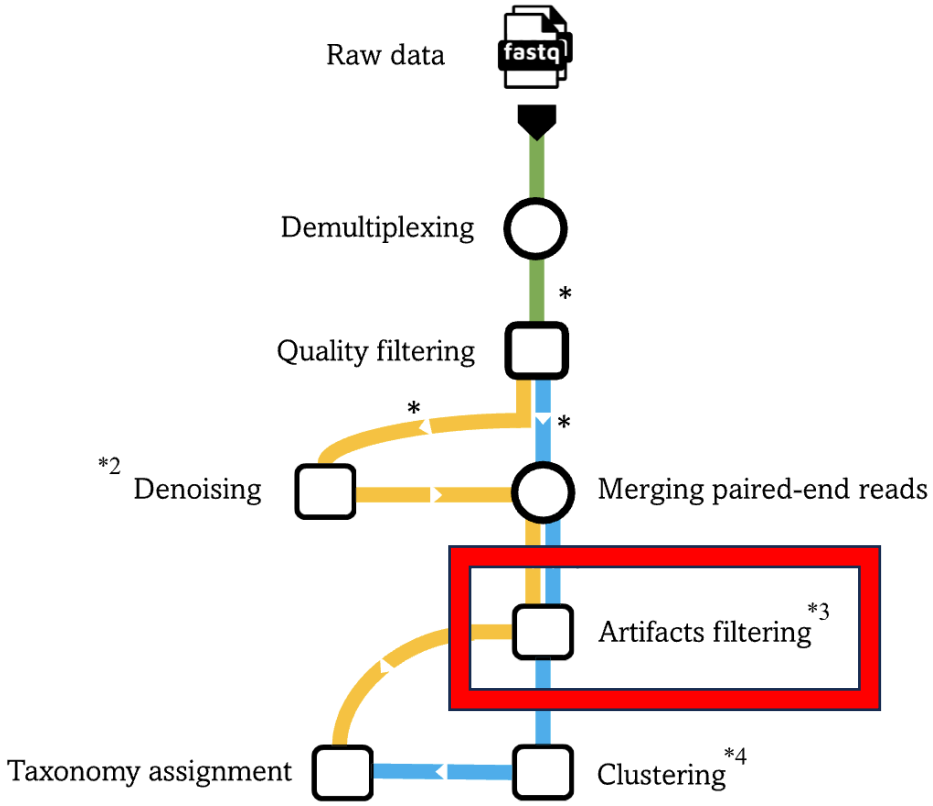
Hakimzadeh et al 2024

Correcting for sequencing errors.
DADA2, UNOISE

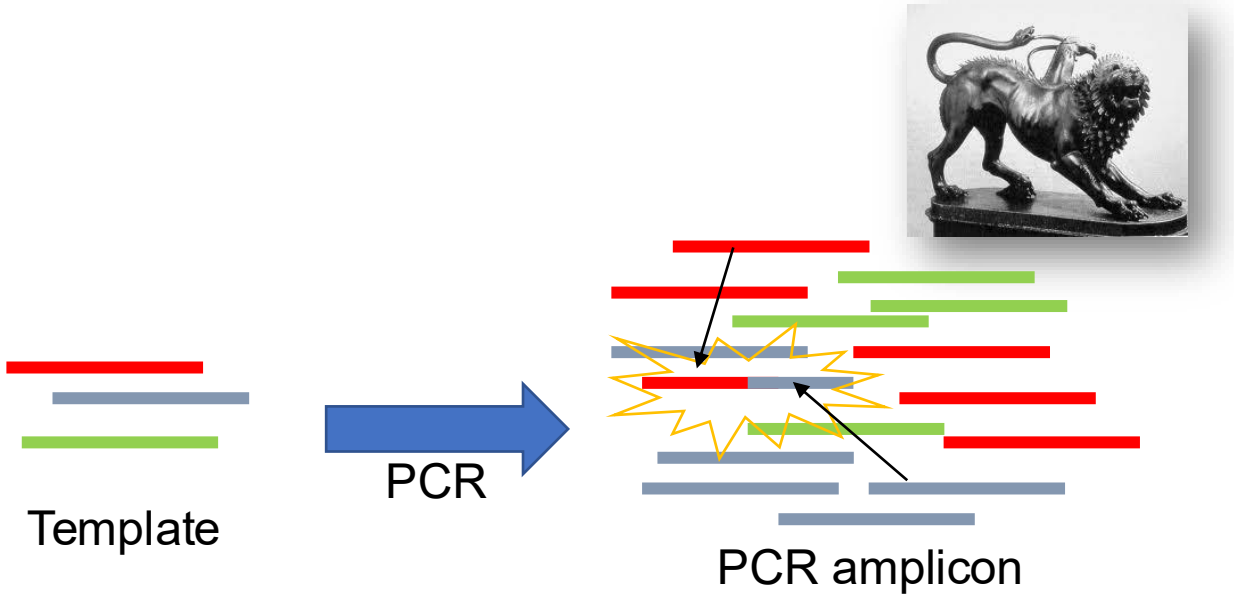


ATCGGGTCGTTGCTATTGC
ATCGGGTCGTTACTATTGC

Metabarcoding: Artifacts filtering (chimera)



Hakimzadeh et al 2024

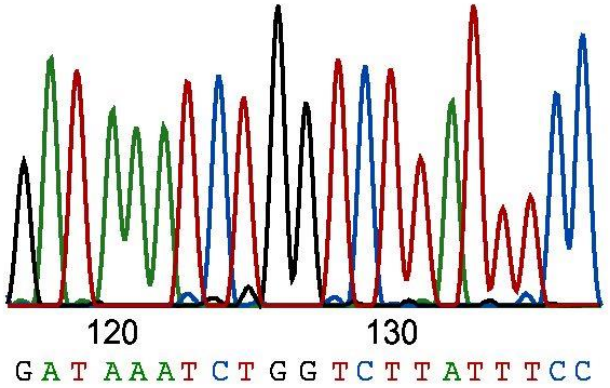


De novo versus reference based chimera checking

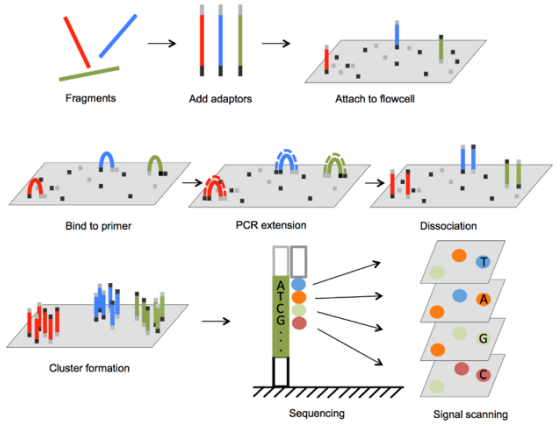
The level of chimeric sequences depends on how variable the marker is! → Be aware of false positives in the tests

PCR-induced errors

- PCR mutations: polymerase enzymes introduce erroneous nucleotides now and then, even those enzymes with proof-reading activity
  - Dependent on the technology whether these become «visible»
  - In classic (direct) Sanger sequencing such errors become «diluted»
  - In methods where your final sequences are derived from one single DNA template, they become visible and must be corrected for!

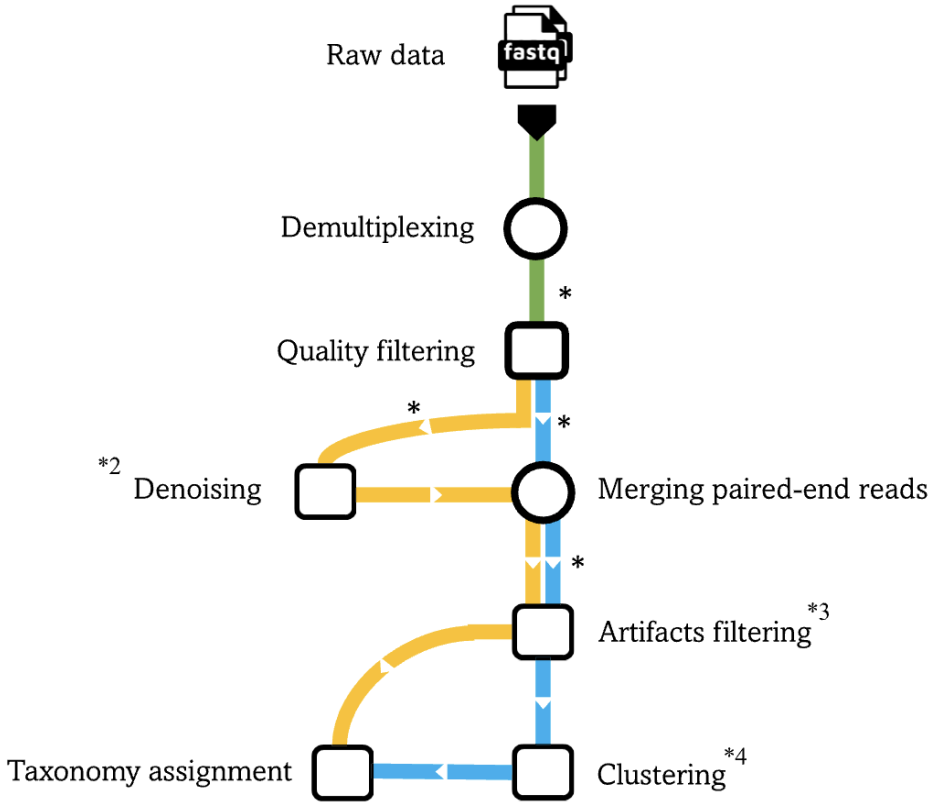


Sanger



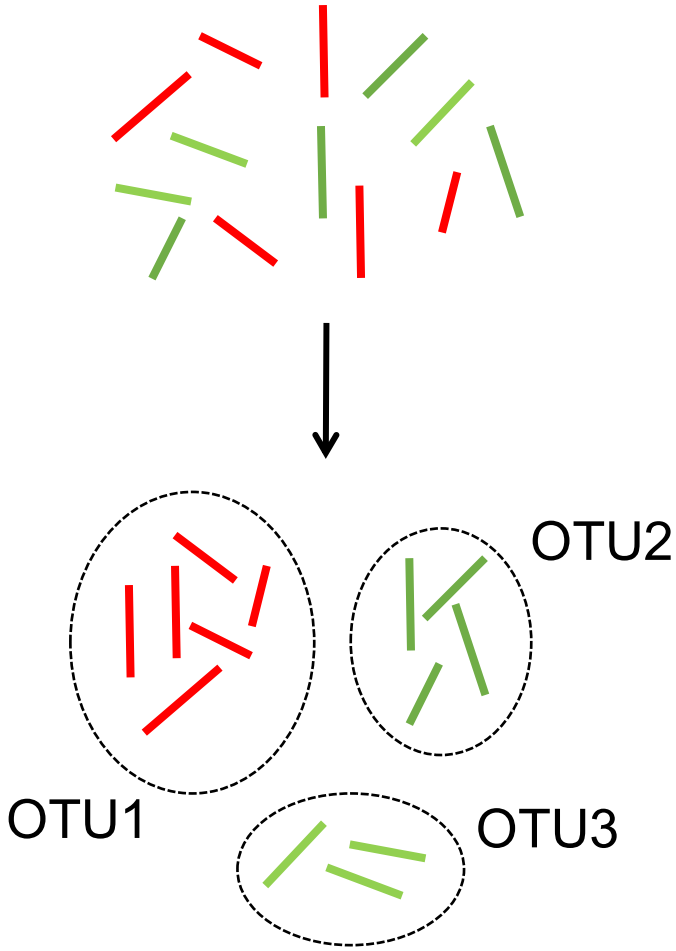
Illumina

Metabarcoding: Clustering

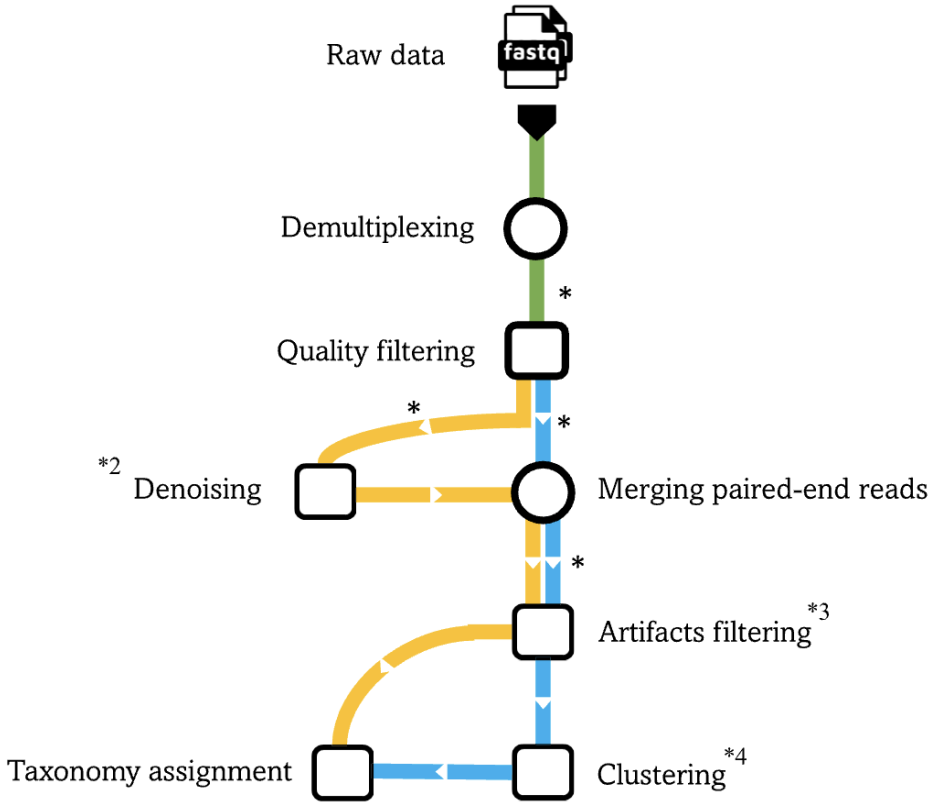


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More on this later!!

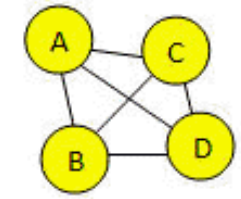


Metabarcoding: clustering

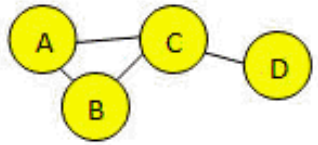


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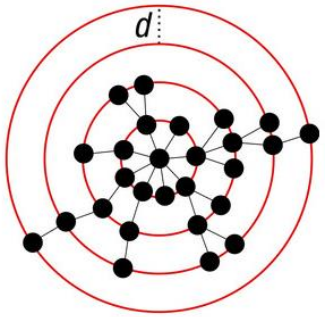
Many different clustering approaches

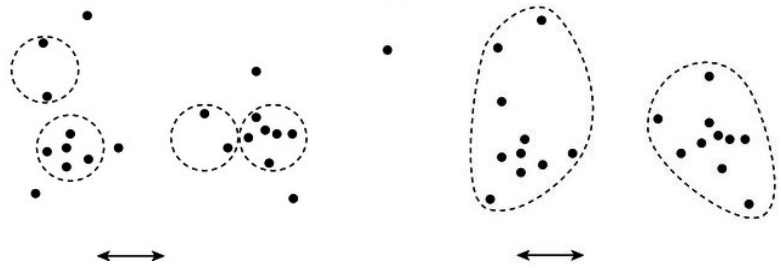


Maximum distance
Complete linkage

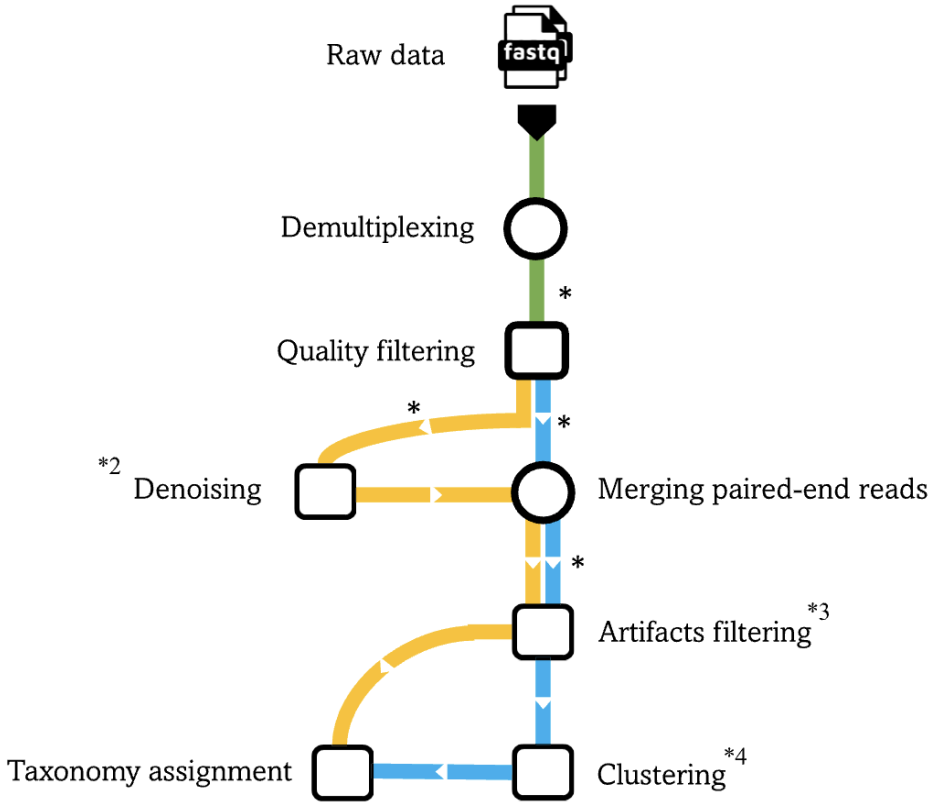


Minimum distance
Single linkage

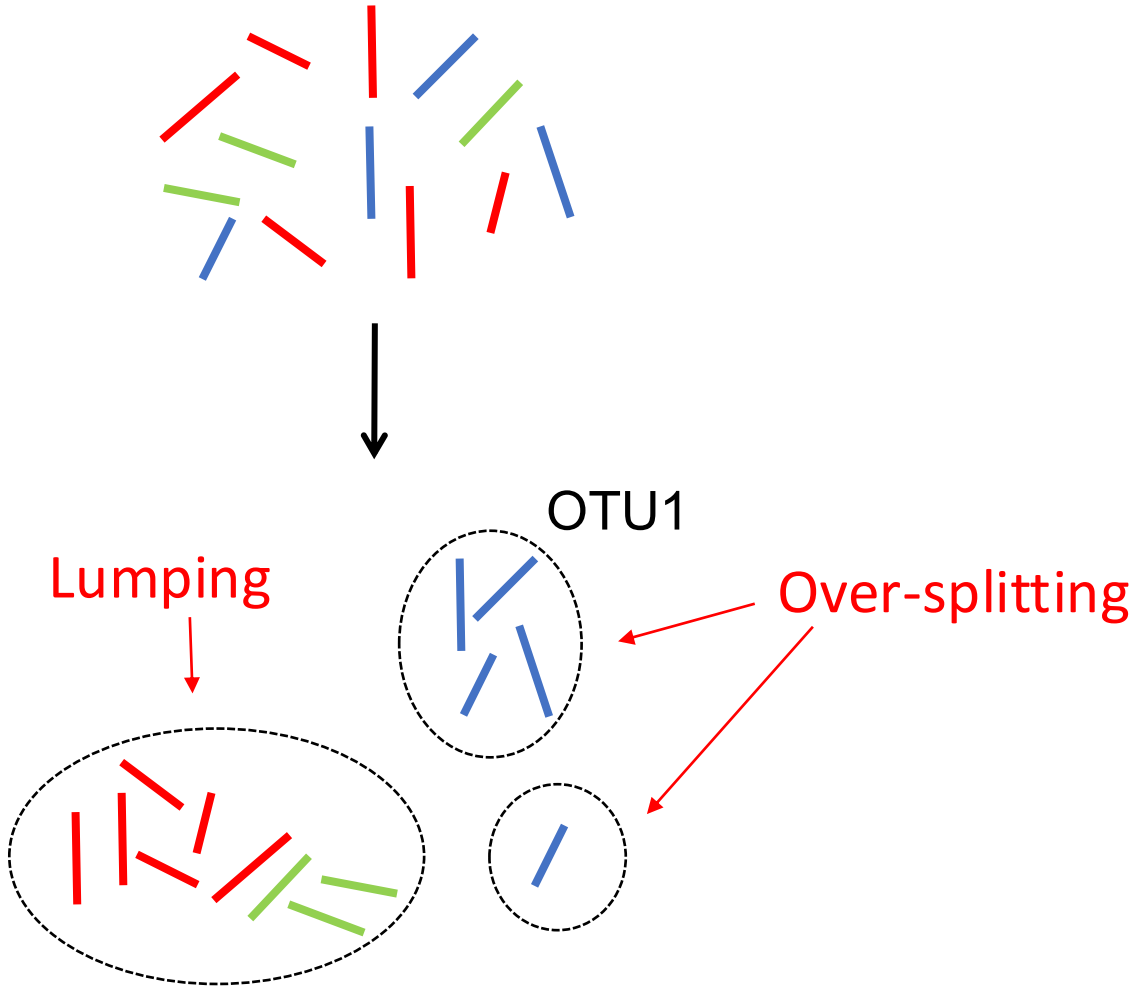




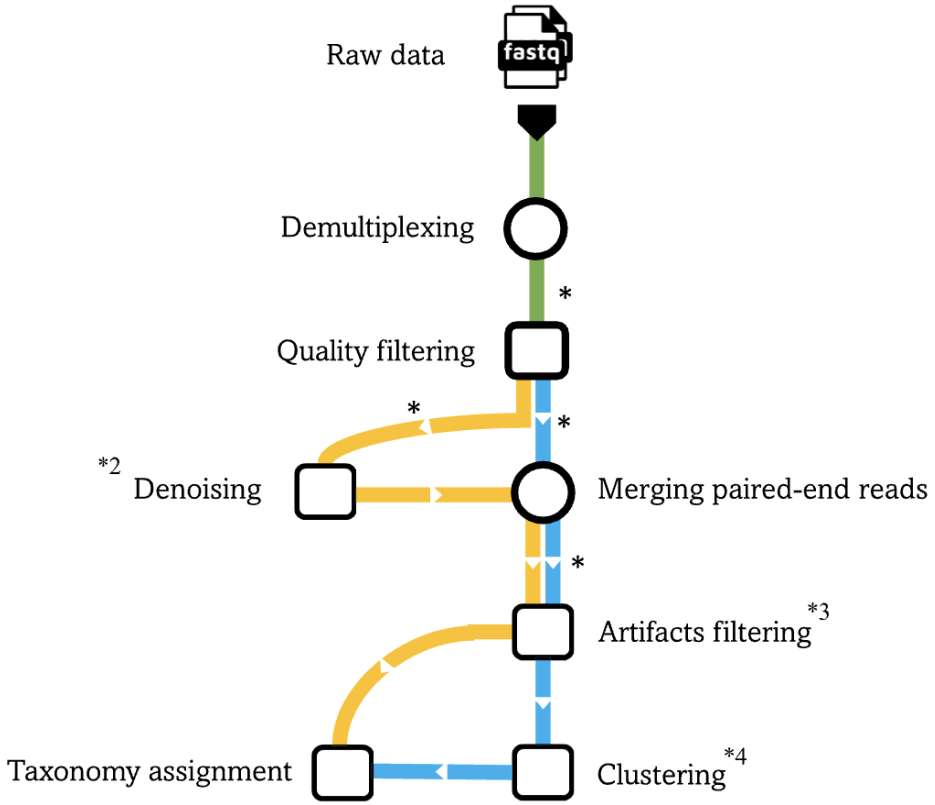
Metabarcoding: Clustering



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Metabarcoding: taxonomic assignment



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unite

PR²

EUK REF

BOLD SYSTEMS

silva
high quality ribosomal RNA databases

Molecular Ecology Resources

RESOURCE ARTICLE OPEN ACCESS

The Ribosomal Operon Database: A Full-Length rDNA Operon Database Derived From Genome Assemblies

Anders K. Krabberød | Embla Stokke | Ella Thoen | Inger Skrede | Håvard Kauserud

Why do taxonomic assignment?

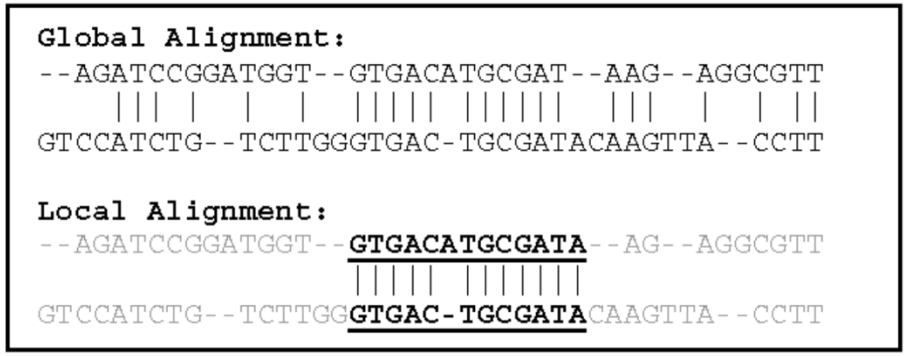
- Not strictly necessary to answer alpha and beta diversity questions
  - Detecting shifts in community composition and genetic diversity doesn't require taxonomic assignments
- Assigning taxonomy links sequences to a wealth of pre-existing information
  - Linking sequences to species improves interpretation and explanation of patterns in alpha and beta diversity
- Choice of marker can impact taxonomic assignment
  - No marker is perfect
  - Discriminating power varies between markers and taxonomic groups
  - Database quality, availability, and completeness varies between markers

Taxonomic assignment - databases

- Different algorithms for comparing sequences to databases and scoring the results
- Alignment based
  - **BLAST**, VSEARCH, OBItools
- Phylogenetic based
  - HmmUFOtu, TIPP, DECARD, SAP
- Kmer-based machine learning approaches
  - RDP, UTAX, SINTAX

Taxonomic assignment - Alignment based

- Alignment based
  - **BLAST (Basic Local Alignment Search Tool) , vsearch**
- Local alignments begin by checking a small piece of the query sequence against the reference database and then expanding the match to find areas of high similarity
- Global alignments find the best match in the reference database across the entire length of the query sequence
- Output is typically an alignment score, percent identity, and coverage score



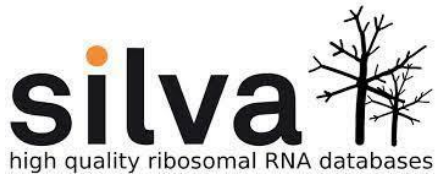
62% similarity

93% similarity
32% coverage

Taxonomic assignment - databases

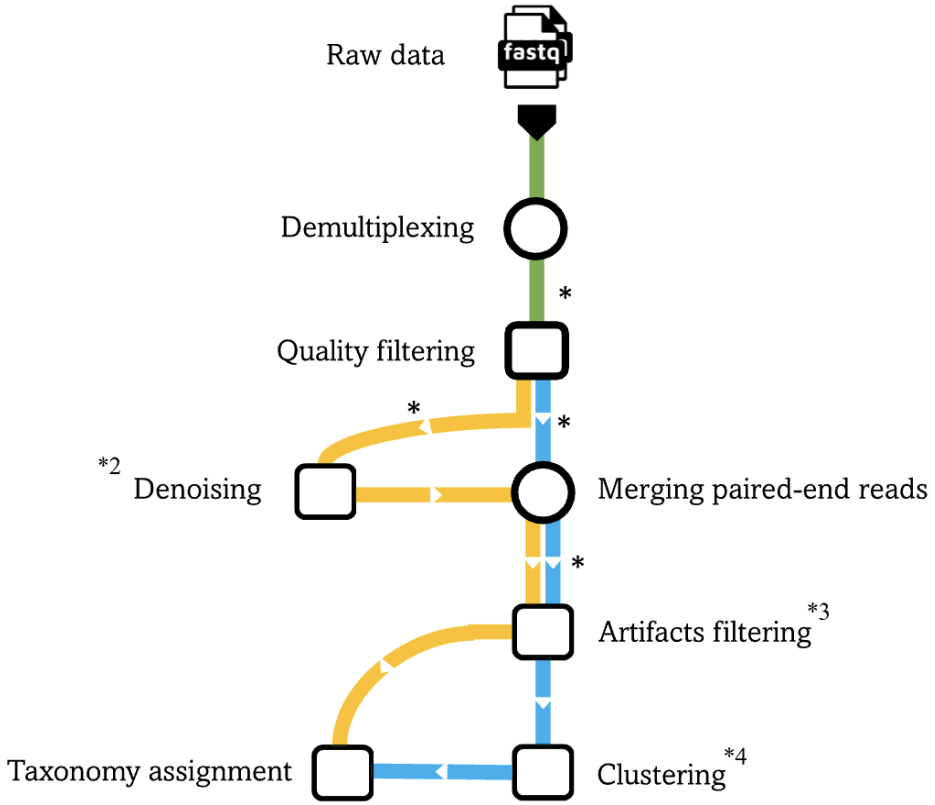
- Taxonomic assignment quality is highly dependent on database accuracy and completeness
  - Misidentified sequences create identification errors and low-quality assignments
  - Missing reference sequences reduce the resolution of taxonomic assignments or result in misidentifications
- Taxonomic assignment is only as good as the database!
  - **PR2**: a curated database for protists 18S
  - **Silva**: 18S and 28S, Bacteria, Archaea, and Eukaryota. Comprehensive, but not curated.
  - **UNITE:** ITS. Mostly used for Fungi, other eukaryotes are also included
  - **ROD:** Ribosomal Operon Database, the full rDNA operon from genome projects
- Quality of taxonomic assignments can often be improved by creating custom-curated reference databases







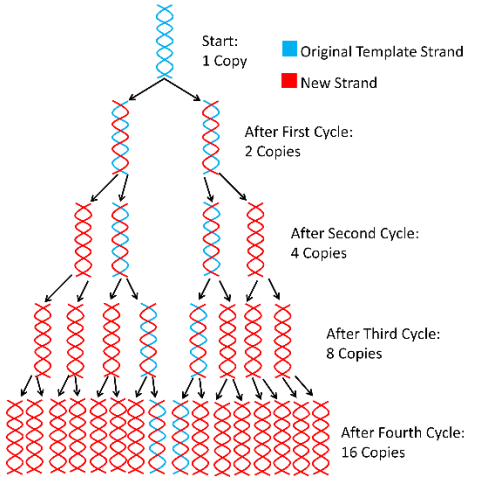
Metabarcoding bioinformatics workflow



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Samples

	1	2	3	4	5	6	7
OTU1	0	0	0	0	0	0	0
OTU2	2	0	10000	0	0	5	0
OTU3	0	0	0	0	0	0	0
OTU4	0	0	0	0	0	0	0
OTU5	0	0	0	0	0	0	0
OTU6	0	500	0	0	0	4	0
OTU7	0	0	0	0	0	0	0
OTU8	0	0	0	0	0	0	0
OTU9	0	0	23		0	30000	0
OTU10	0	0	0	0	0	0	0



Be careful with resampling and transformations!

Check the effect from various data treatments options on the results!

Depends on the study aims!

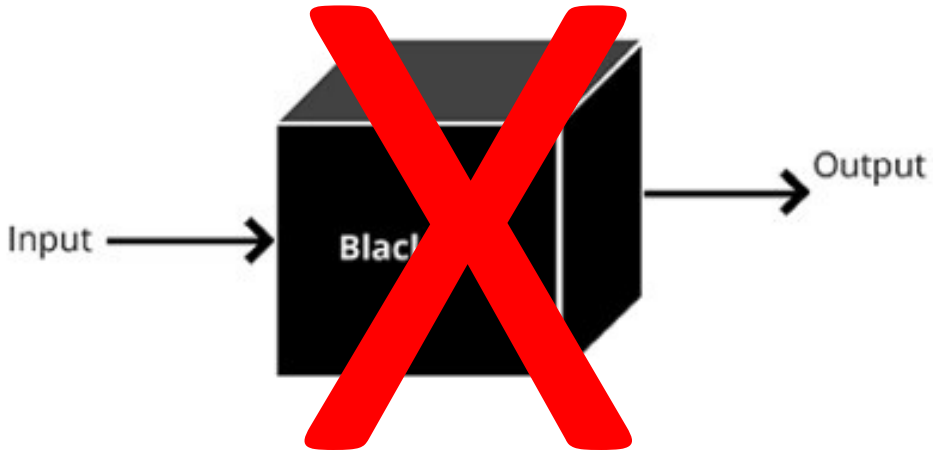
- There are many pipelines and programs to choose from.
- Some have been around a long time and is well maintained Qiime, mothur, VSEARCH, DADA2
- Or you can pick different tools for different parts of the workflow to better suit your data.

Pipelines	First/latest release	Software type	Feature	Demux	Primer/adapter removal	Marker
AMPtk	Nov 2017/Jan 2023	Precompiled	ASV, OTU	Yes	Yes	16S, 28S, COI, ITS
Anacapa	June 2018/June 2018	Precompiled	ASV	Multilocus demux based on primers	Yes	Multimarker
APSCALE	Jan 2022/Feb 2023	Precompiled	ASV, OTU	No	Yes	Multimarker
Barque	Sep 2016/Dec 2022	Precompiled	ASV, OTU	No	Yes	Multimarker
BIOCOM-PIPE	Aug 2020/Jan 2021	Precompiled	ASV, OTU	Yes	Yes	16S, 18S, 23S
Cascabel	Nov 2020/Dec 2022	Precompiled	ASV, OTU, swarm-cluster	Yes	Yes	Multimarker
CoMA	Dec 2020/Jan 2022	Precompiled	ASV, OTU, swarm-cluster	No	Yes	Multimarker
DADA2	May 2016/Nov 2022	Set of algorithms	ASV	No	Yes	Multimarker
Dadaist2	May 2020/Jul 2022	Precompiled	ASV	No	Yes	Multimarker
dadasnake	Apr 2020/Feb 2023	Precompiled	ASV, OTU	No	Yes	Multimarker (with ITSx)
DAnIEL	Apr 2021/May 2021	Precompiled	ASV	Yes	Yes	ITS
eDNAflow	Jul 2020/Jan 2021	Precompiled	ASV	Yes	Yes	Multimarker
FROGS	Jan 2016/Apr 2023	Set of algorithms	ASV, swarm-cluster	Yes	Yes	Multimarker (with ITSx)
gDAT	Mar 2020/Jan 2021	Precompiled and set of algorithms	OTU	No	Yes	18S, ITS
JAMP	Jan 2017/Jan 2022	Precompiled	OTU	Yes	Yes	Multimarker (without taxonomic assignment)
LotuS2	May 2021/Apr 2023	Precompiled	ASV, OTU, swarm-cluster	Yes	Yes	Multimarker (16S, 18S, 23S, 28S, ITS with ITSx)
MetaWorks	Jun 2020/Mar 2023	Precompiled	ASV, OTU	No	Yes	Multimarker; SSU (12S, 16S, 18S), ITS (with ITSx), LSU (28S), COI (with pseudogene removal), rbcL (with pseudogene removal)
MICCA	Oct 2014/Aug 2018	Precompiled	ASV, OTU, swarm-cluster	Yes	Yes	16S, 18S, 28S, ITS
mothur	Feb 2009/May 2022	Set of algorithms	ASV, OTU	Yes	No	Multimarker
NextITS	July 2022/May 2023	Precompiled	ASV, OTU, swarm-cluster	Yes	Yes	ITS (with ITSx)
nf-core/ ampliseq	Dec 2018/Mar 2022	Precompiled	ASV	No	Yes	Multimarker (with ITSx)
OBITools3	Sep 2019/Sep 2022	Set of algorithms	ASV	Yes	Yes	Multimarker
PEMA	Feb 2020/Dec 2021	Precompiled	OTU, swarm-cluster	No	Yes	Multimarker
PipeCraft2	Dec 2021/Dec 2022	Precompiled and set of algorithms	ASV, OTU	Yes	Yes	Multimarker (with ITSx and pseudogene removal)
PIPITS	Dec 2014/Nov 2022	Precompiled	OTU	No	Yes	ITS (with ITSx)
QIIME 2	July 2016/May 2023	Set of algorithms	ASV, OTU	Yes	Yes	Multimarker (with ITSxpress)
SCATA	Feb 2010/Oct 2021	Precompiled	OTU	Yes	Yes	Multimarker (with ITSx)
SEED2	Feb 2018/Oct 2020	Precompiled	OTU	Yes	Yes	16S, ITS (with ITSx)
Tourmaline	July 2022/Apr 2023	Precompiled	ASV, OTU	Yes	Yes	Multimarker
USEARCH	2010/Mar 2020	Set of algorithms	ASV, OTU	No	No	Multimarker
VSEARCH	Nov 2014/Sep 2022	Set of algorithms	ASV, OTU	No	No	Multimarker
VTAM	Oct 2020/May 2022	Precompiled	ASV, OTU	Yes	Yes	Multimarker (with pseudogene removal)

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Which methods to use???

- Which methods to use? → No general answer – it is context-dependent. You must argue for your choices!



Biases and sources of errors

- **Extraction bias.** No universal DNA or RNA extraction method will work for all organisms and environments.
- **Marker bias**. DNA and RNA markers differ substantially in length, taxonomic resolution, copy number, and alignability.
- **Primer bias.** Primers differ in their melting temperature, binding specificity, and coverage of the targeted organisms.
- **PCR bias.** Differential amplification, mistakes in nucleotide incorporation, and chimaera formation; *keep amplification cycles low.*
- **Sequencing bias.** Differences depend strongly on the platform and sequencing strategy.
- **Bioinformatics biases.** Incorrect classification, false positive, or invalid operational taxonomic units (OTUs) formation
- **Poor clustering**. Accumulation of sequencing errors leads to wrongly constructed OTUs, which can lead to overestimation of richness. Use stringent denoising and removal of rare OTUs
- **Unequal sequencing depth**. Having different numbers of sequences between samples complicates tests of taxonomic diversity because OTU accumulation curves fail to reach a plateau in large composite samples.

Nilsson et al. 2019